

Chapter 2: Multidimensional Data Visualization

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Introduction to Multi-Dimensional Data

- **Multi-dimensional data** is often given as a **table**
 - Each item (data point) has a number of attributes
 - Attributes can be categorical (nominal or ordinal), or numerical (discrete or continuous)
 - Dimensionality refers to the number of attributes per item
- Often, multi-dimensional data does not have a “natural” spatial layout. *Examples:*
 - Specification of cars (MPG, horse power, brand...)
 - Health care situation (infant mortality, number of physicians per citizen, money spent on health care...)
 - Time series of stock rates

Strategies: Multidimensional Data Vis

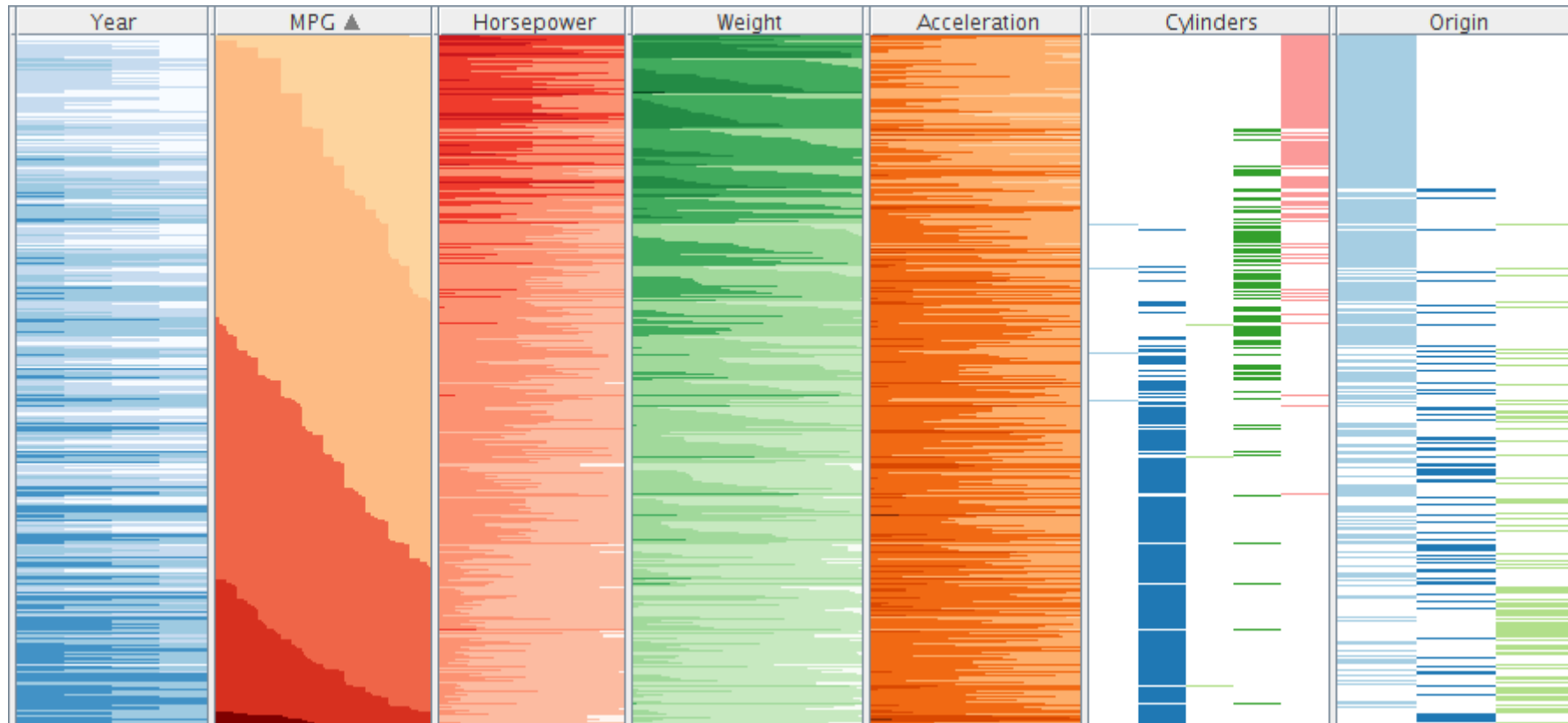
- 2-3 attributes can be mapped to space
- Intermediate number ($p \approx 3-10$) of attributes can be visualized using multidimensional visualization techniques
 - Topic of today's lecture
 - We will see one example with >100 attributes
- Very high number typically requires dimensionality reduction
 - Topic of Chapter 3

Section 2.1:

Basic Techniques

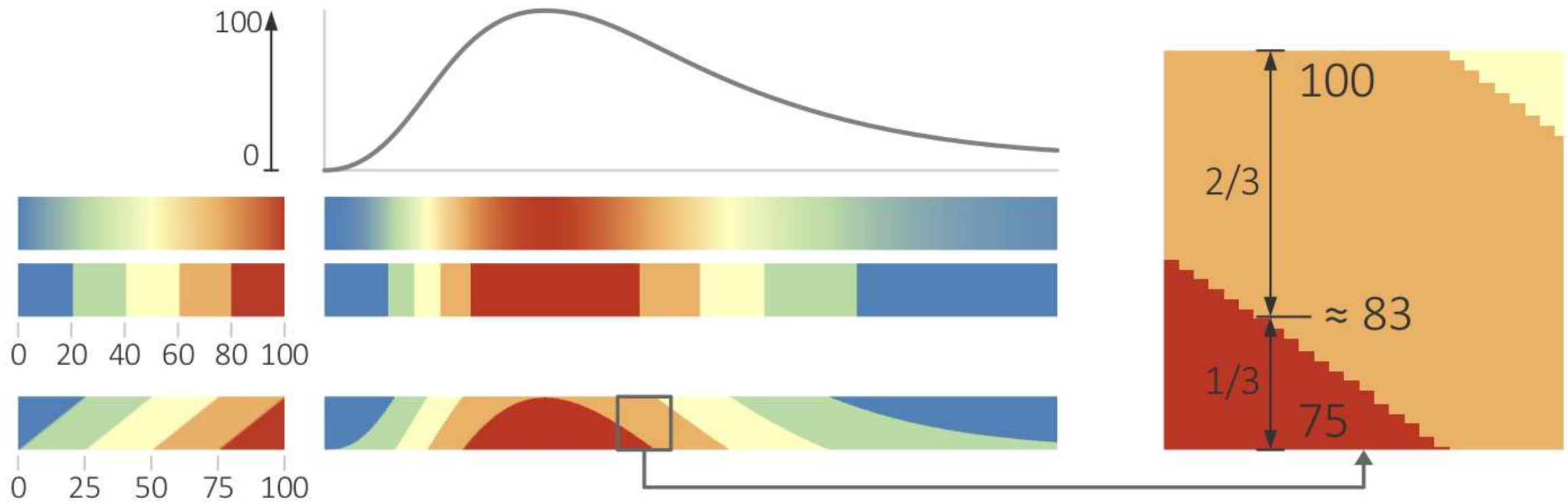
Table-based Visualization

- *Simple idea:* Keep table format, visualize numerical and categorical values with color codes



Two-Tone Coloring

- **Two-tone coloring** is a space-efficient way to encode values with high read-off accuracy



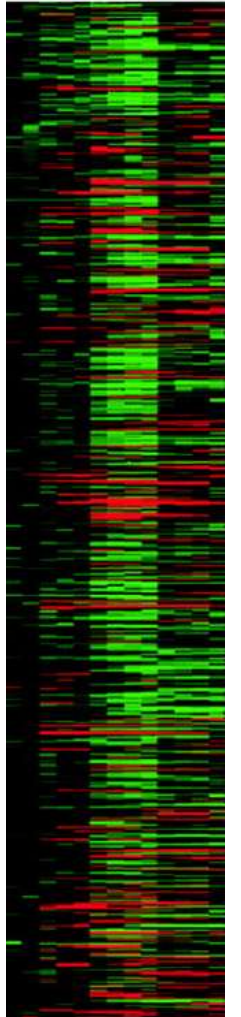
Interaction Technique: Table Lens

- Table-based visualizations are meant to provide an overview. Interaction techniques such as the **table lens** add back details (in the form of text labels) on demand.



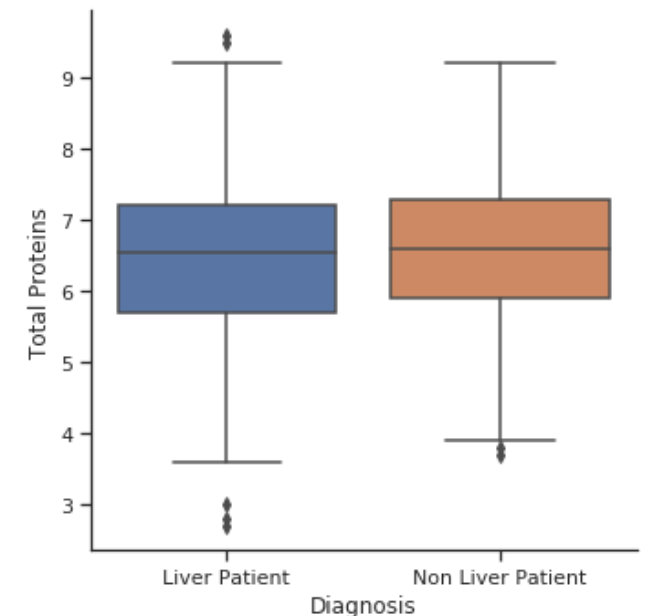
Heat Maps

- In some applications, color coded tables are referred to as **heat maps**
 - Heat maps typically only contain numerical values, and use the same color coding scheme for all cells
 - Example [Eisen et al. 1998]:
 - Rows = Different genes
 - Columns = Different times after treatment
 - Values = log of relative expression level
 - Diverging color scheme for increase/reduction in expression



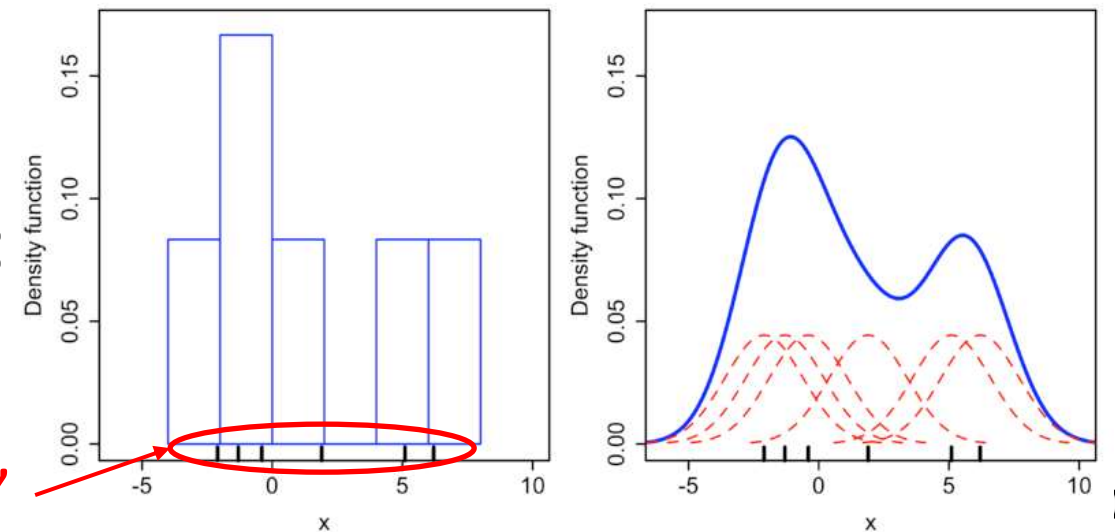
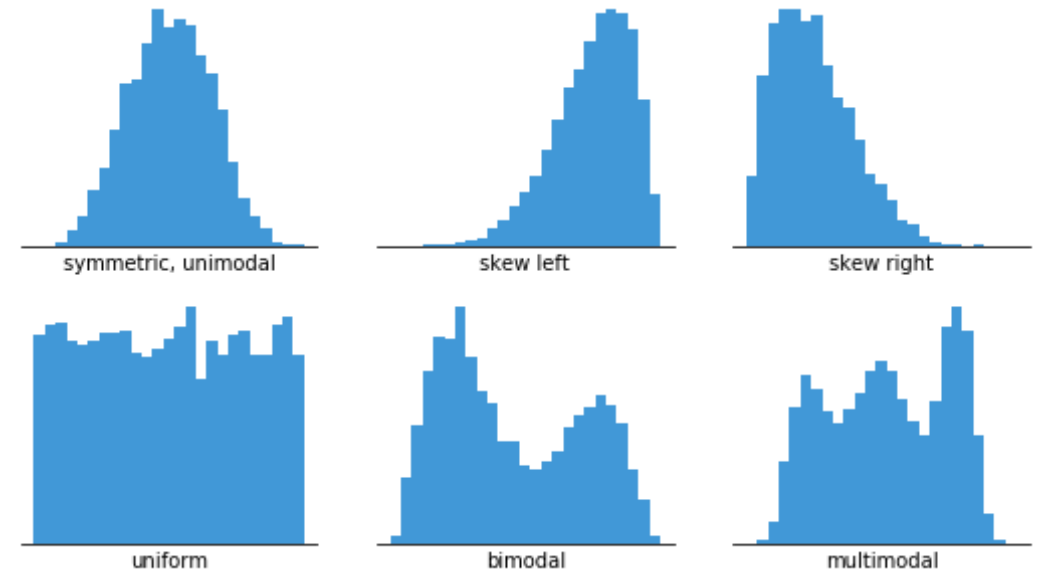
Box Plots

- **Box plots** (box-and-whisker plots) summarize the distribution of a single attribute
 - Different variants exist; one common definition:
 - Central band represents median
 - Box indicates interquartile range (IQR), i.e., it extends from the 1st to 3rd quartile = 25th to 75th percentile
 - Whiskers indicate remaining points up to 1.5 IQRs of the 1st/3rd quartile
 - Other points drawn individually as outliers
 - Specify convention in legend!
 - Placing boxplots side-by-side reveals extent of overlap between distributions



Histograms and Kernel Density Estimation

- **Histogram:** Bar chart of number (or fraction) of values $\mathbf{x}_i$ in predefined ranges (“bins”)
 - Bin width is a relevant parameter
 - If $\mathbf{x}_i$ are random samples, histogram is a piecewise constant estimate of the corresponding probability density function (PDF)
- **Kernel Density Estimation (KDE):**
 - Smooth estimate of the PDF



“rug plot”

KDE: Formal Definition

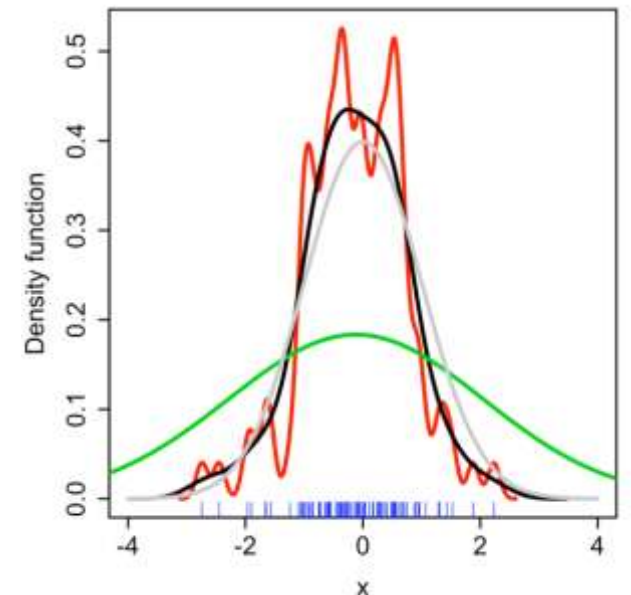
- Given values $\mathbf{x}_i$, KDE is defined by

$$f(\mathbf{x}) = \frac{1}{n} \sum_i K(\mathbf{x} - \mathbf{x}_i)$$

- Kernel K is normalized to integrate to unity, e.g., Gaussian

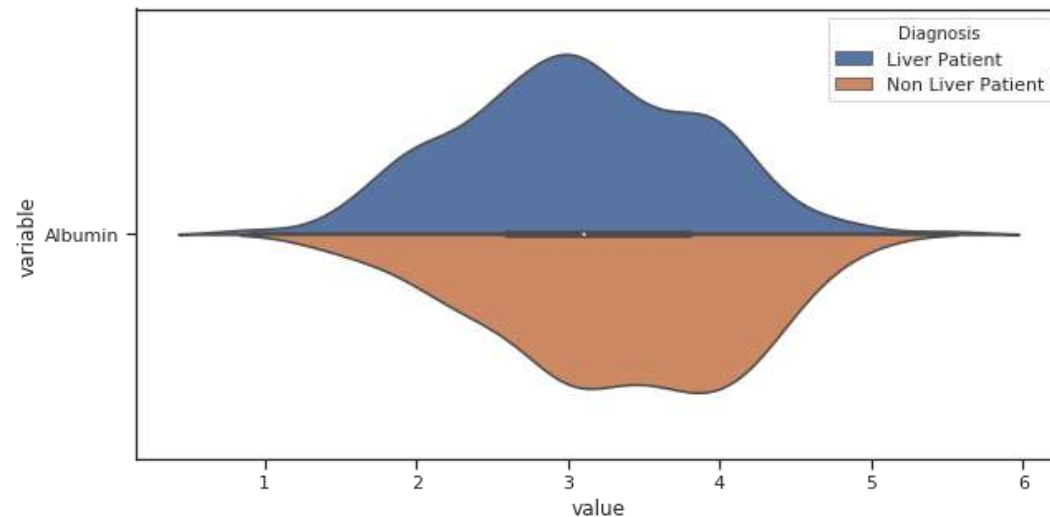
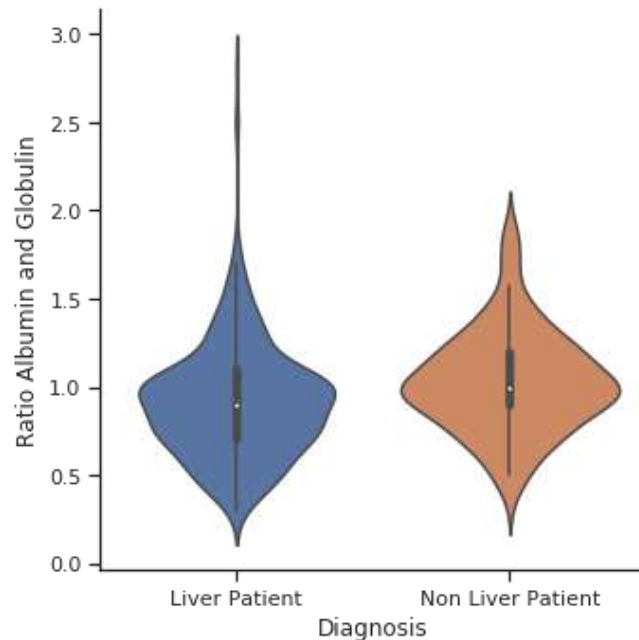
$$K(x) = \frac{1}{h\sqrt{2\pi}} e^{-\frac{x^2}{2h^2}}$$

- Bandwidth h** controls smoothness of f :
 - Analogous to number of bins in histogram
 - h too small: plot “rough”/noisy
 - h too large: plot overly smooth
 - h should decrease with larger number n of samples
 - Scott’s rule of thumb for 1D Gaussians: $h \approx 1.06 \frac{\hat{\sigma}}{\sqrt[5]{n}}$



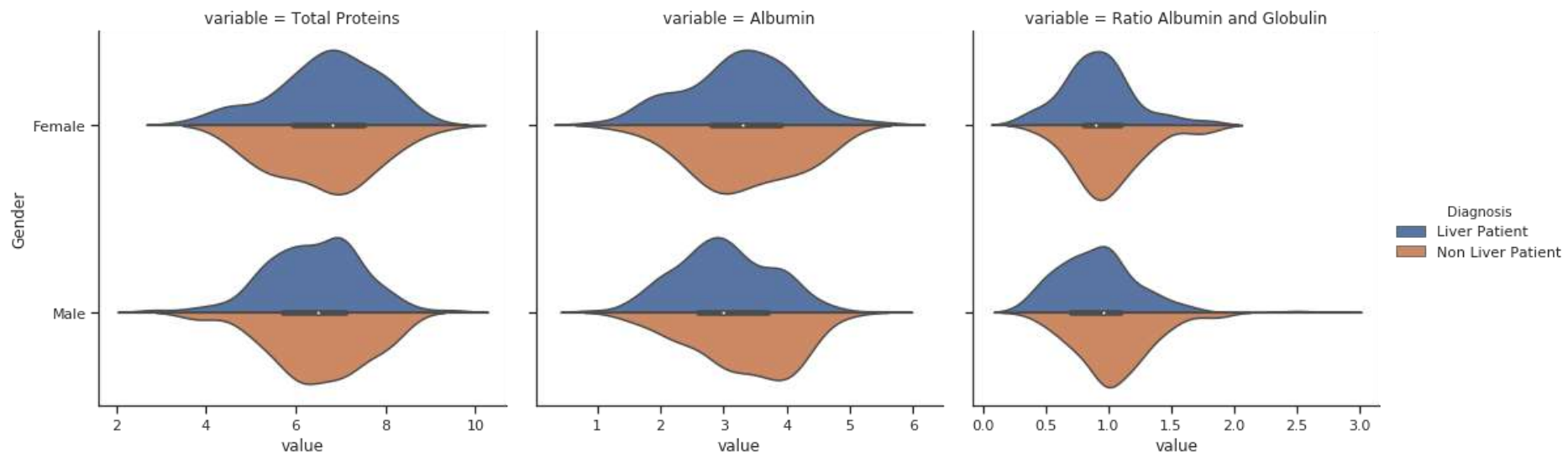
Violin Plots

- **Violin plots** combine boxplots with a kernel density estimate
 - More detailed impression of distribution
 - Split violin plots can be used for comparison
 - *But:* Depend on kernel and bandwidth



Faceting

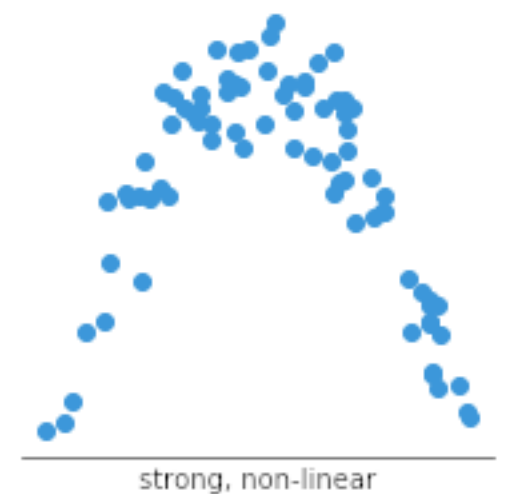
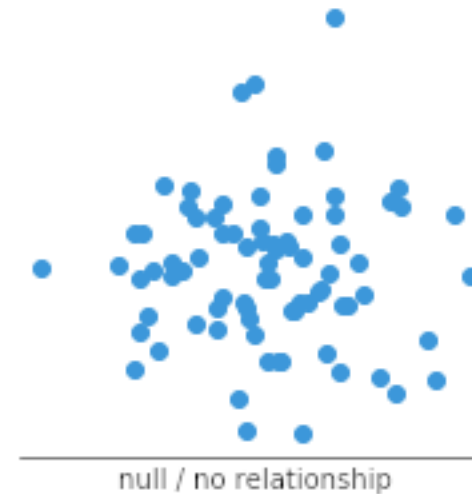
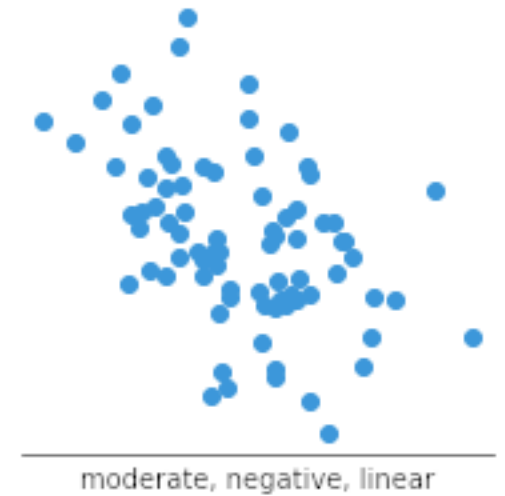
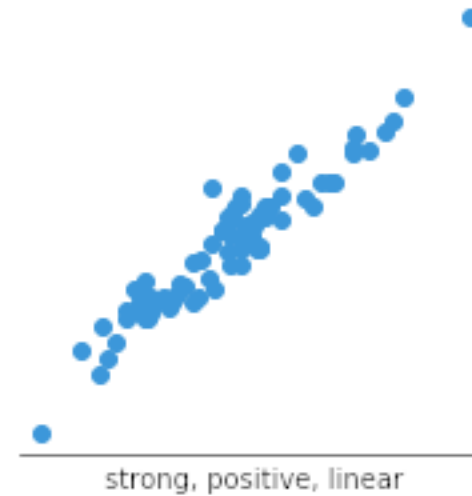
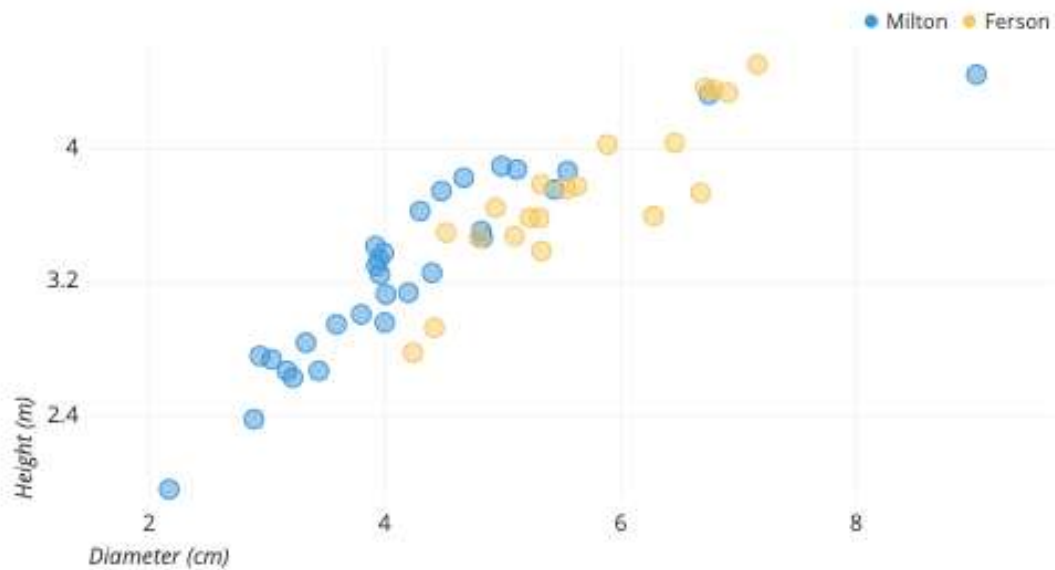
- **Caveat:** Avoid overloading individual plots!
 - e.g., color + marker types + line style + line width...
- *Alternative:* Grid of multiple plots, each of which shows some “facet” of the data



Section 2.2: Scatterplots

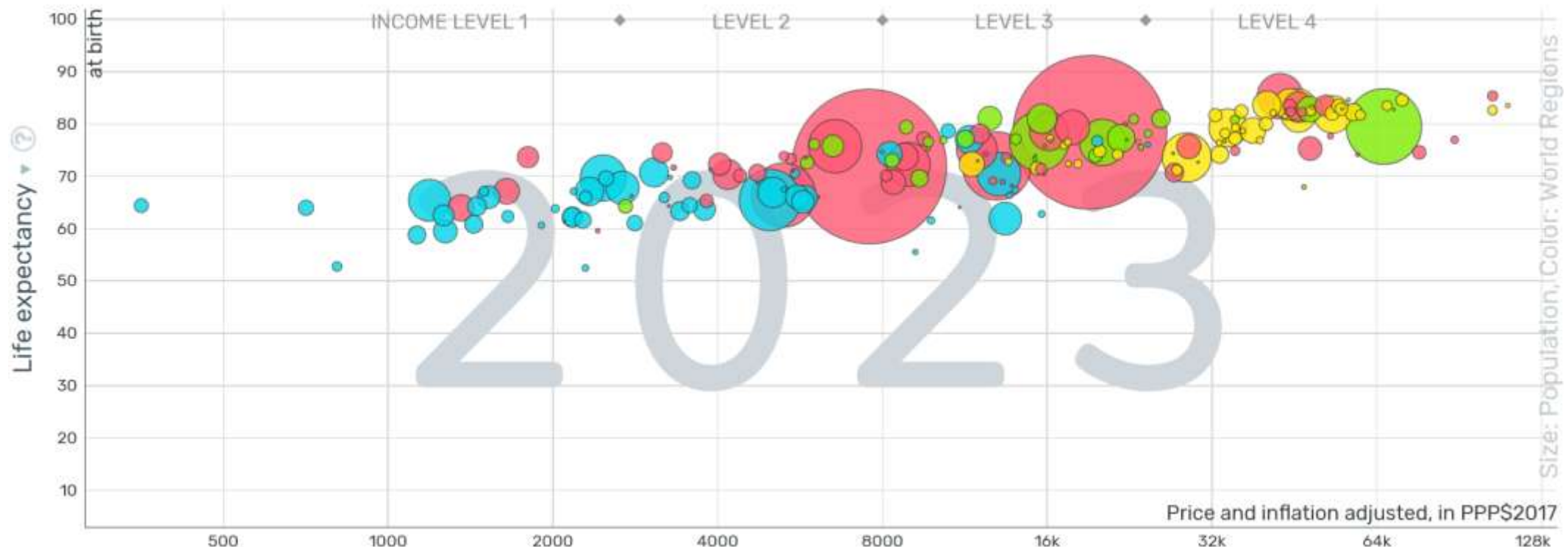
Scatterplots

- **Scatter plots** convey the relationship between two variables
 - Items represented as small markers, positioned according to the two values
 - Additional variables can be encoded via marker size, shape, color



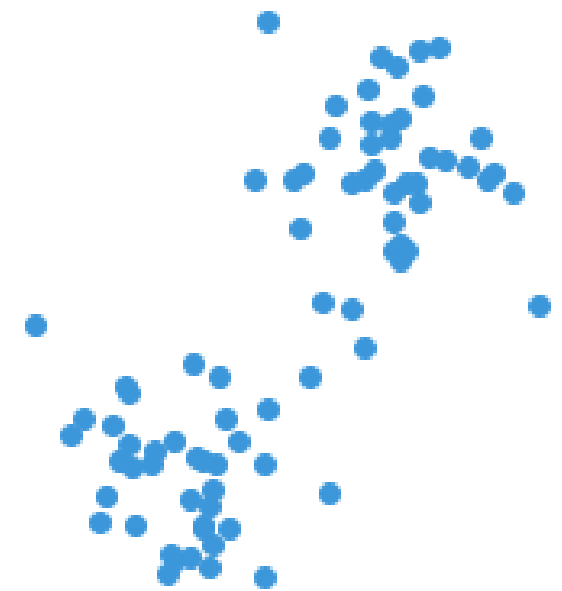
Aside: Bubble Charts

- **Bubble charts** are scatterplots in which markers are scaled according to a numerical attribute
 - Attribute should be proportional to area, not radius!
 - Increases overload, make markers semi-transparent!

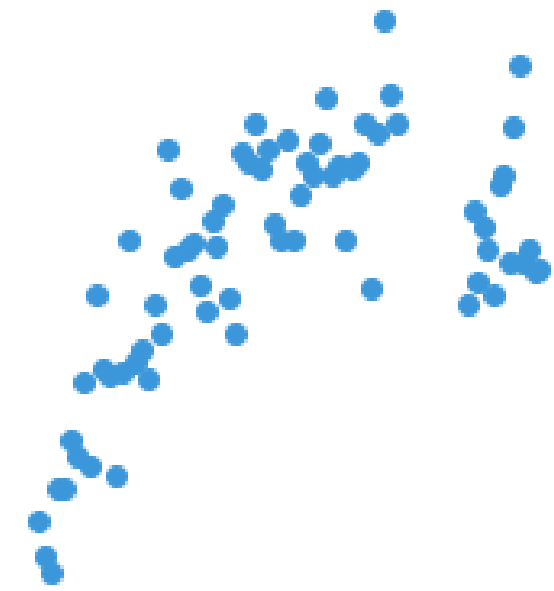


Scatterplots: Clusters, Gaps, Outliers

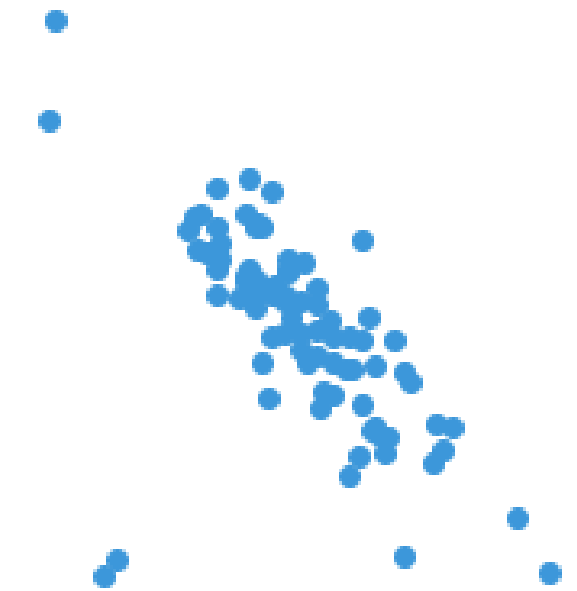
- **Scatter plots** also reveal **clusters**, **gaps**, and **outliers** in the data



data clusters (two)



gap in values



outliers (lower left)

Limitation: Overdraw

- Too many data points lead to **overdraw**
- **Partial solutions:**
 - Change size or shape of markers
 - Subsample the data
 - Reduce opacity of markers
 - Color code a 2D histogram or Kernel Density Estimate (“heat map”)
 - *Disadvantage:* No longer see individual points

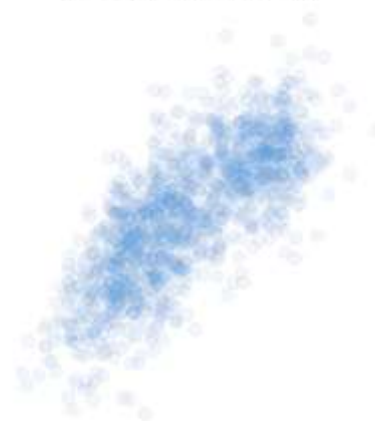
Original data, 1500 points



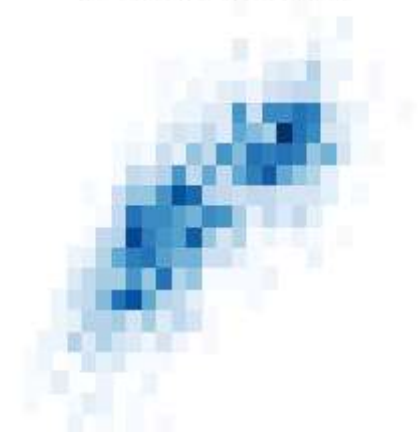
Sampled data, 400 points



Plot w/ Transparency

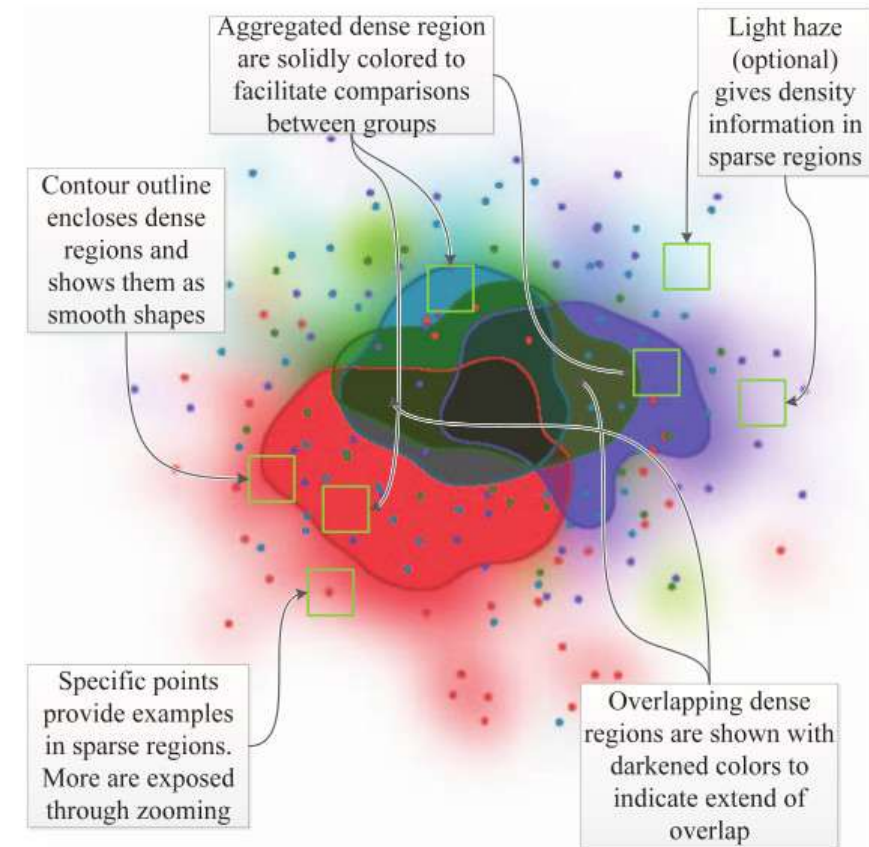
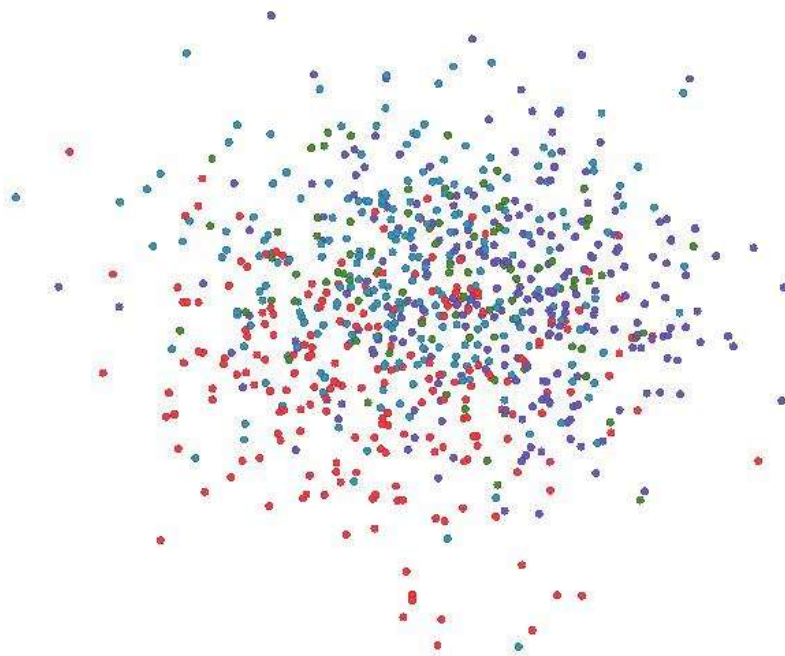


Plot as 2-d histogram



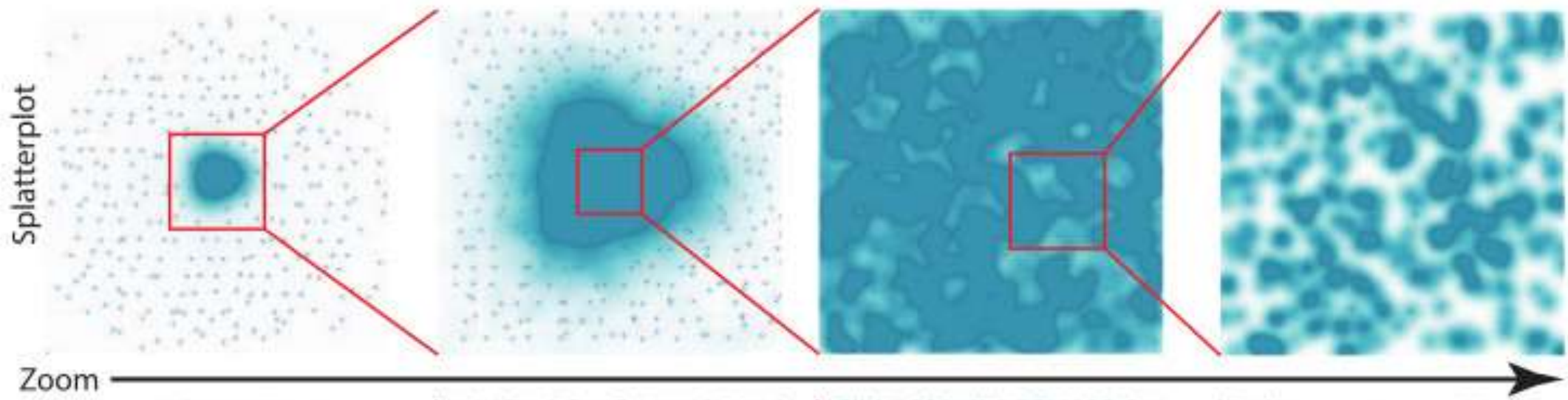
Splatterplots

- Splatterplots [Mayorga / Gleicher 2013]
 - Visual abstraction in image space, detail added when zooming in



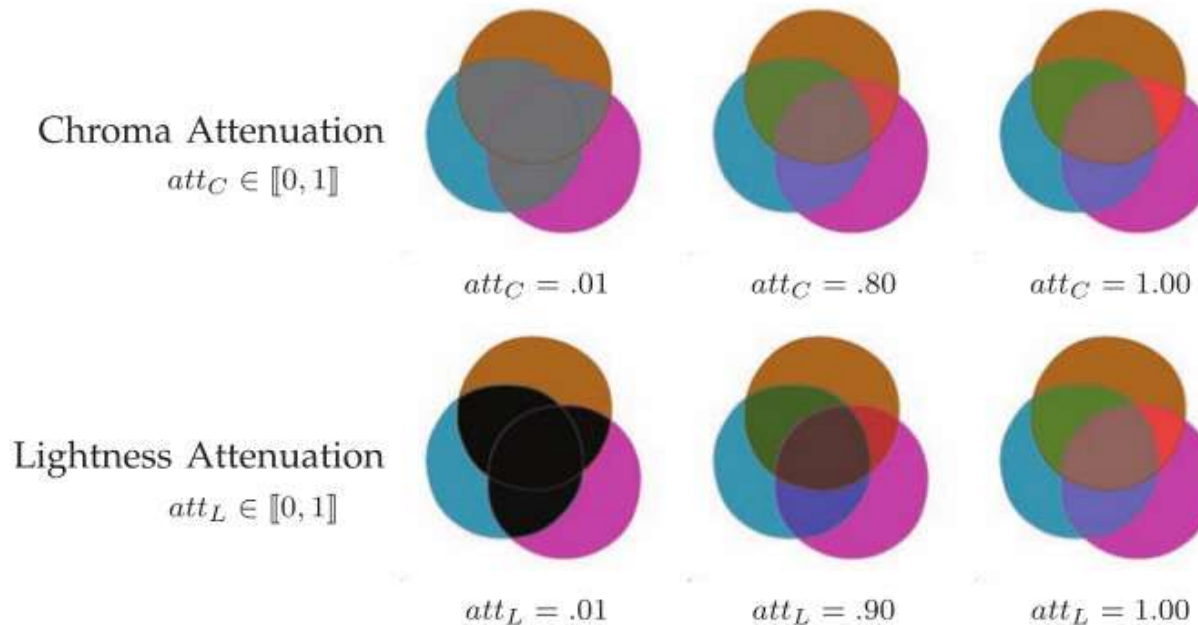
Splatterplots: Dense Regions

- Based on **Kernel Density Estimation**
 - Kernel width defined *in screen space*
 - High-density regions shown as smooth, filled, and bounded shapes
 - *Drawback*: Shape of “dense” regions depend on the chosen threshold



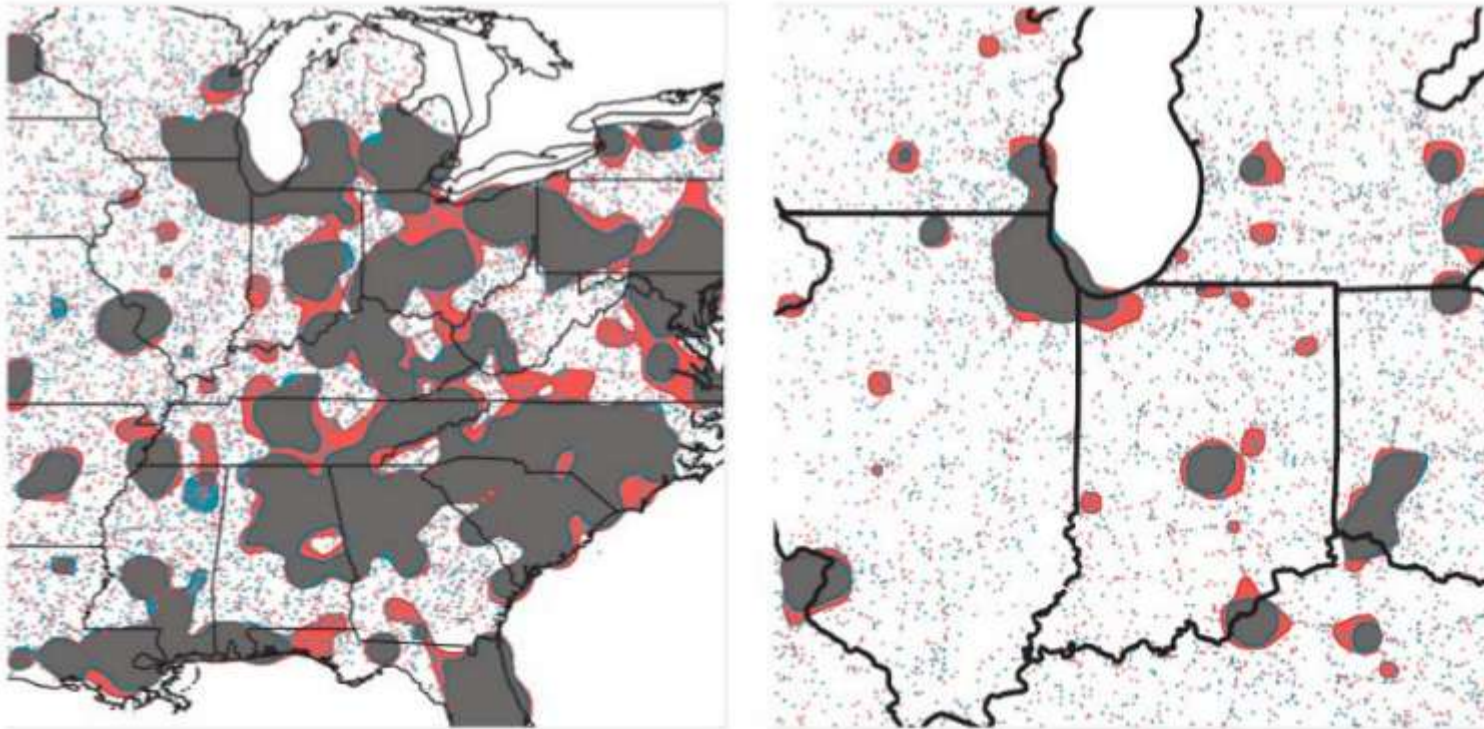
Splatterplots: Color Blending

- Where dense regions overlap:
 - Use hue to encode classes
 - Perform blending in a perceptually uniform color space (Chapter 4)
 - Reduce luminance and saturation to indicate overlap



Splatterplots: Subsampling

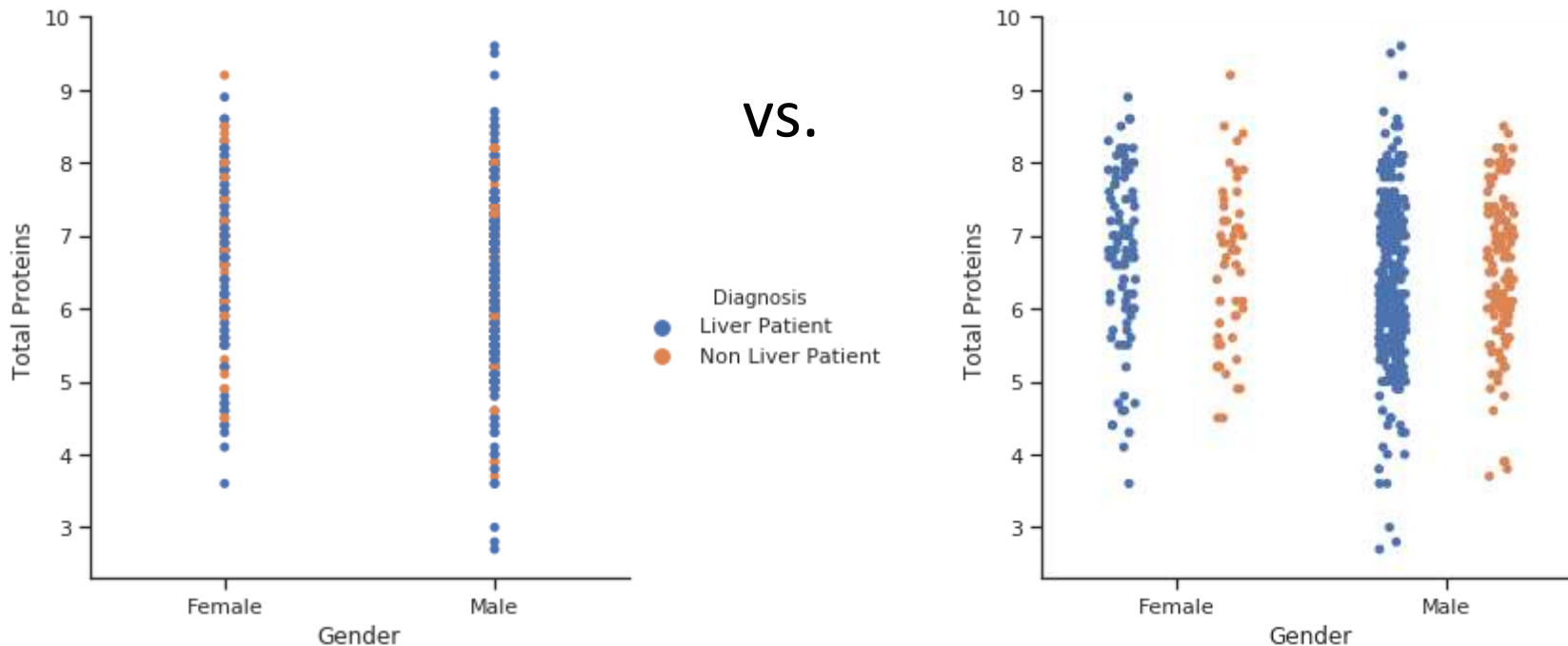
- Outside of dense regions, points are
 - subsampled to ensure a minimum distance between them
 - More points added when zooming in



Fatal car crashes in
2005 vs. 2010

Strip Plots

- Scatter plots in which one attribute is categorical are sometimes called “strip plots”
 - Consider adding offsets and/or random jitter



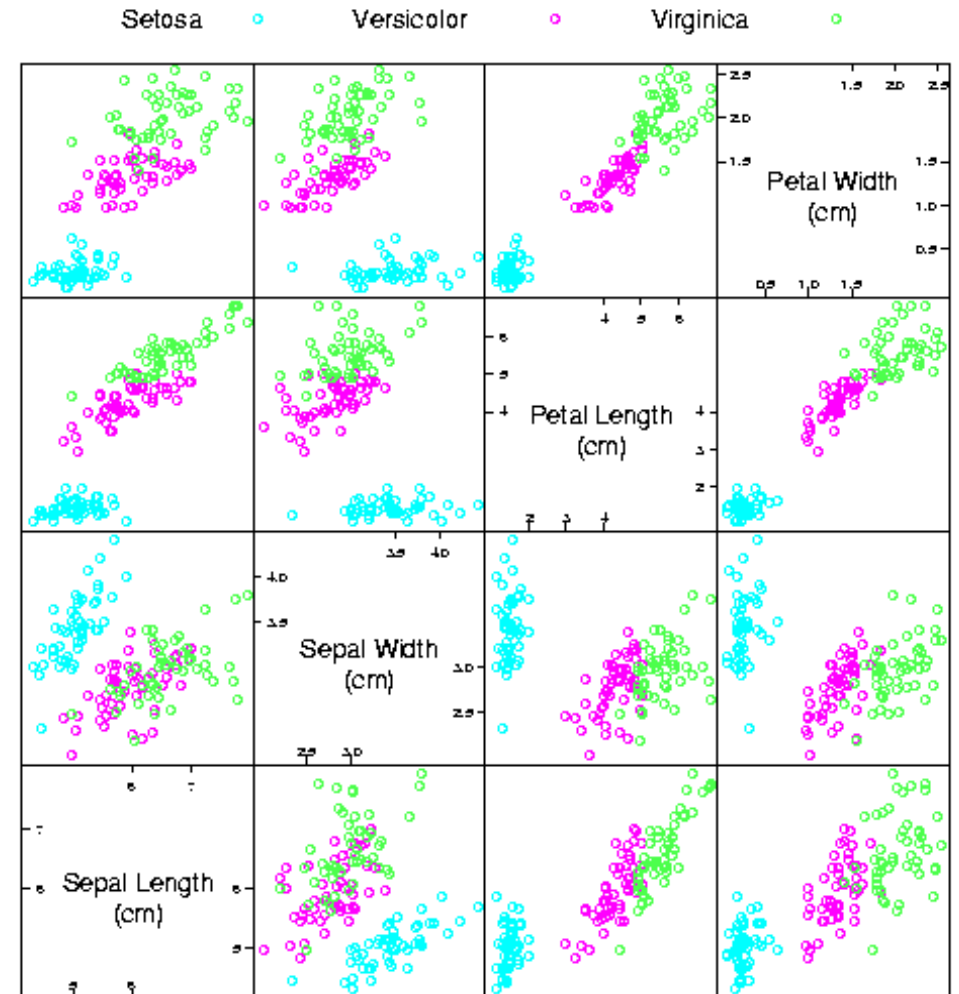
Scatterplot Matrices (SPLOM)

Produce scatterplots for all pairs of variables and place them into a matrix

Total of $(p^2-p)/2$ unique scatterplots

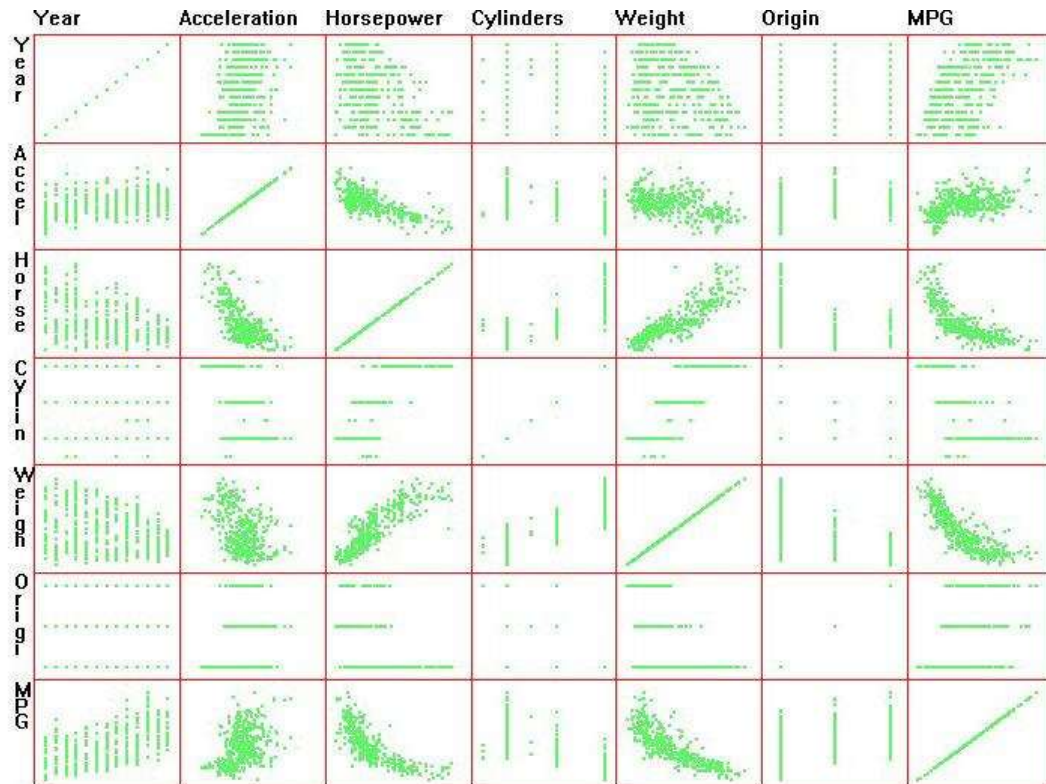
Question: How to use boxes on the diagonal?

Question: Does order matter?

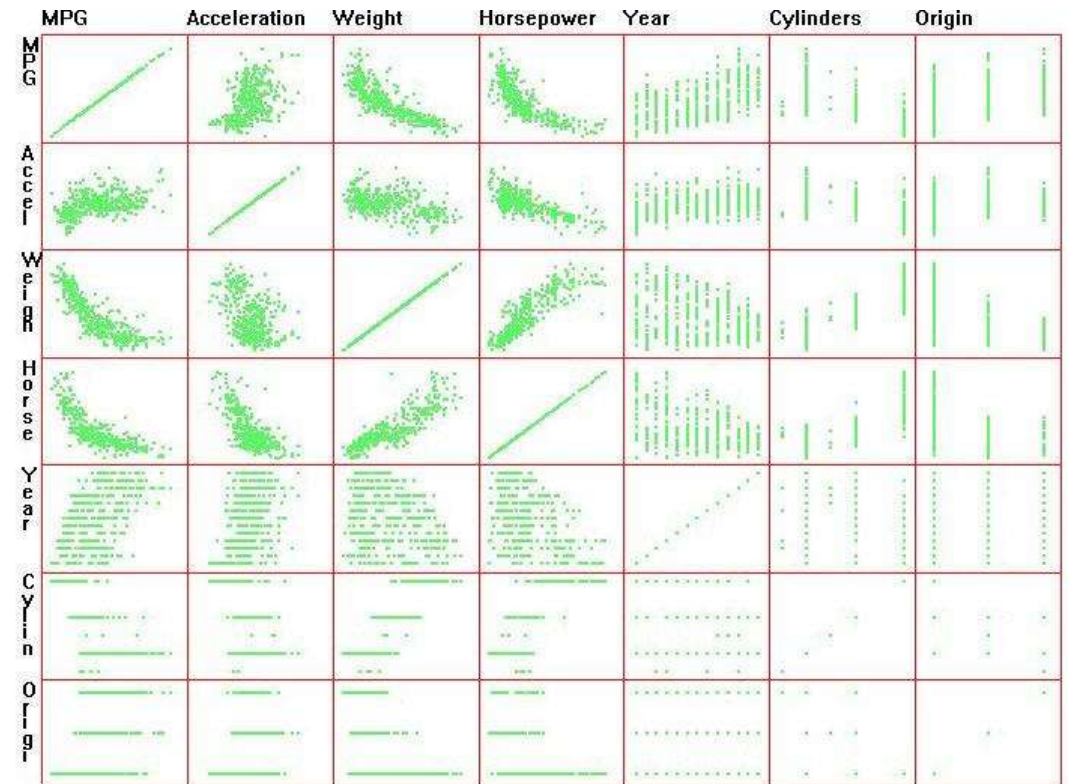


SPLOMs with “Clutter Reduction”

- Re-ordering the dimensions permutes individual plots
- Some layouts are easier to navigate than others



(a)



(b)

Image taken from [Peng et al. 2004]

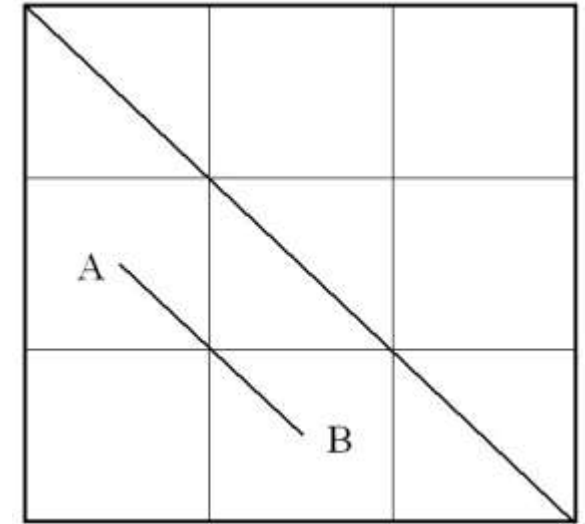
Order in SPLOMs

- [Peng et al. 2004] aim for a “clean” layout in which similar scatterplots are close to each other
 - Distinguish between *high-cardinality dimensions* (number of possible values > number of points) and *low-cardinality dimensions*
 - Sort low-cardinality by number of values
 - Similarity of high-cardinality dimensions based on their correlation
- **Pearson Correlation Coefficient** ρ_{xy} computes the degree of *linear* dependence between **x** and **y** from samples (x_i, y_i) with sample means $\bar{x}, \bar{y}$:

$$\rho_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

SPLOMs: Rating High-Cardinality Order

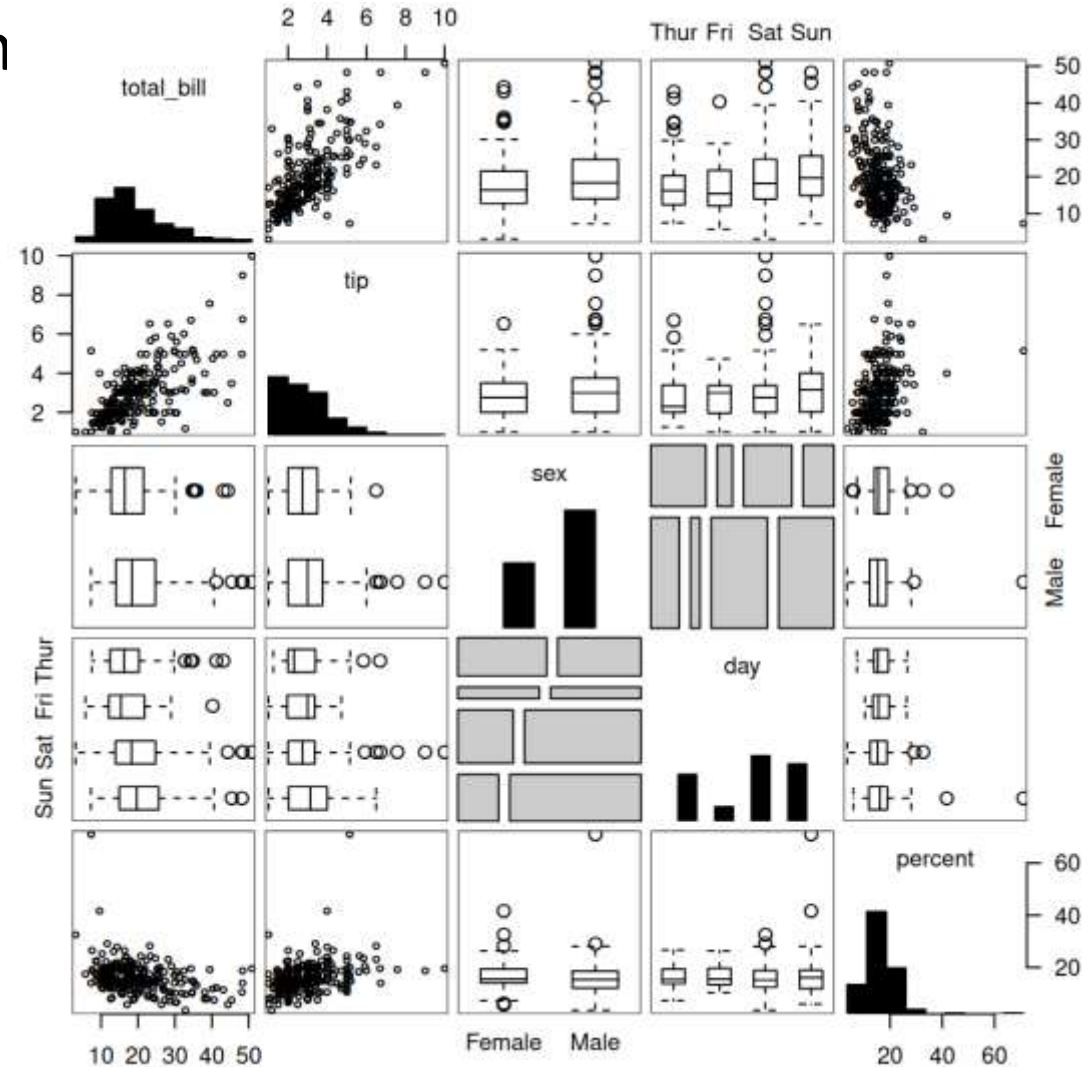
- Peng et al.'s clutter measure for SPLOMs:
 - For each pair (x,y) of high-cardinality dimensions, find all other pairs (x',y') with $|\rho_{xy} - \rho_{x'y'}| < \epsilon$
 - Add the Euclidean distances of all those plots in the SPLOM (the smaller, the better)



- Exhaustive optimization is computationally infeasible
- Heuristic: Random swapping with fixed computational budget

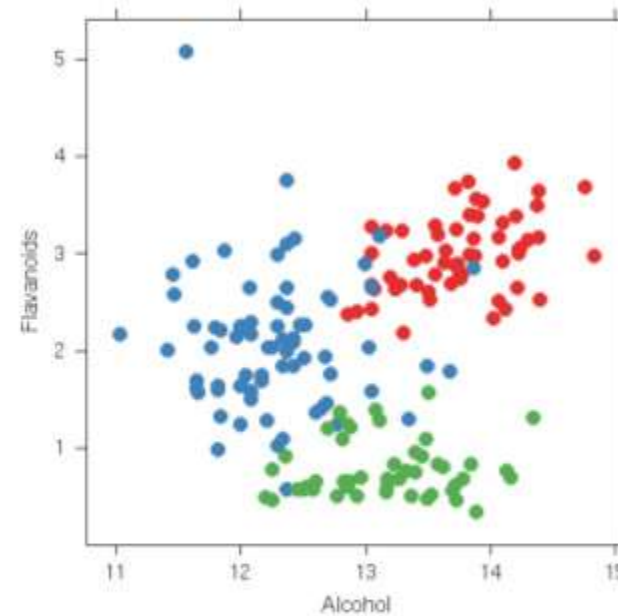
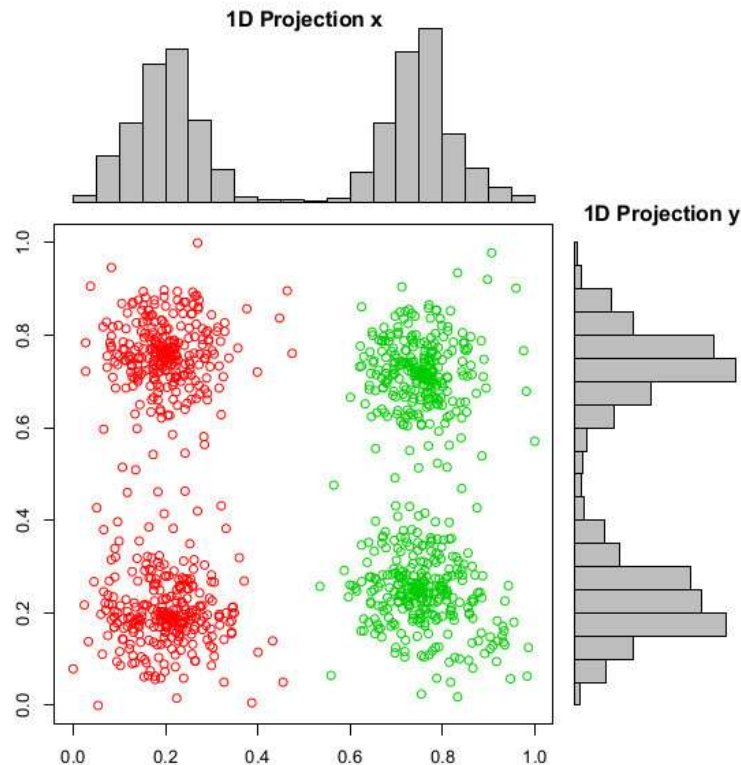
Generalized Pairs Plot

- In a **generalized pairs plot** [Emerson et al. 2012], plot types are selected according to data characteristics.
- Example:
 - *Quantitative-quantitative*: Scatterplot
 - *Quantitative-categorical*: Box plot
 - *Categorical-categorical*: Mosaic plot
 - *Diagonal*: Histogram

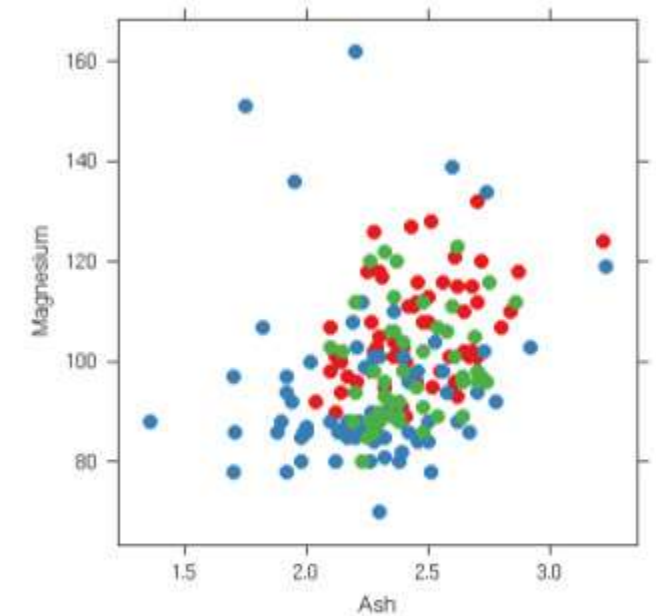


View Selection via Class Separation

- Select scatterplots in which classes are well-separated
 - Labels can be part of the data or obtained by clustering



(a) DSC=90



(b) DSC=49

Distance Consistency

- **Distance consistency (DSC)**

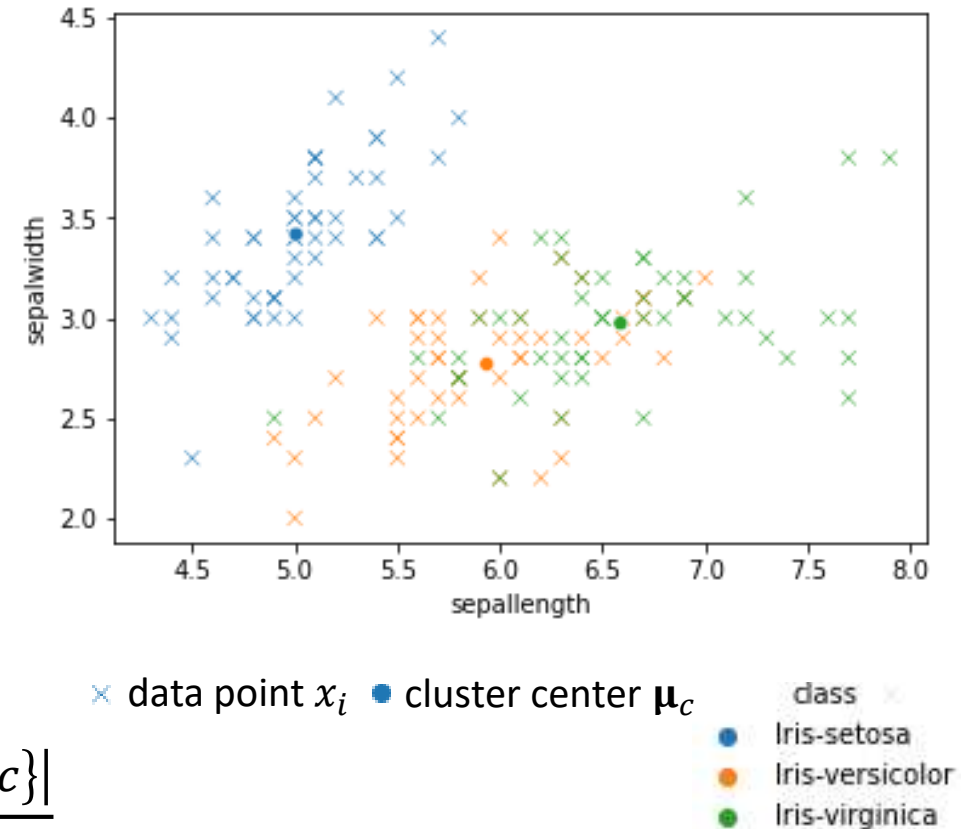
- Defined as the percentage of all points that are closer to their own cluster center than to all others

- Given $|C|$ clusters $c \in C$, each with $|c|$ points $\mathbf{x}_{i \in c}$ in current projection

- Center of cluster c : $\boldsymbol{\mu}_c = \frac{1}{|c|} \sum_{i \in c} \mathbf{x}_i$

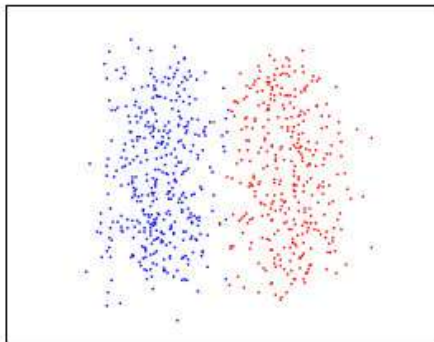
- $$\text{DSC} = 100 \times \frac{\sum_{c \in C} |\{\mathbf{x}_{i \in c} \mid \|\mathbf{x}_i - \boldsymbol{\mu}_c\| < \|\mathbf{x}_i - \boldsymbol{\mu}_{c'}\| \ \forall \ c' \neq c\}|}{\sum_{c \in C} |c|}$$

- Works well when clusters are spherical

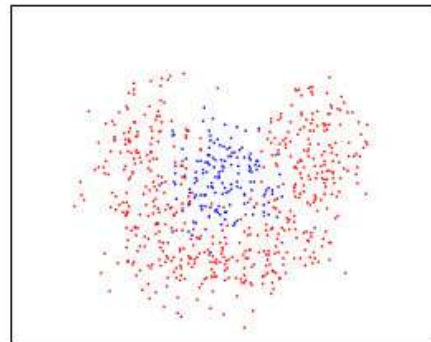


Distribution Consistency

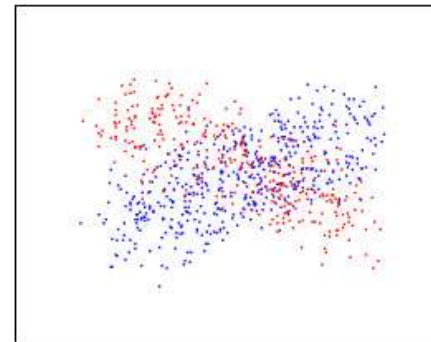
- **Distribution consistency (DC)** penalizes local entropy (amount of uncertainty/mixing of classes) in high-density regions
 1. For each location (x, y) in our plot, estimate per-class density $p_c(x, y)$ and overall $p(x, y)$ using KDE
 2. Compute local entropy: $H(x, y) = -\sum_{c \in C} \frac{p_c(x, y)}{\sum_{c' \in C} p_{c'}(x, y)} \log_2 \left(\frac{p_c(x, y)}{\sum_{c' \in C} p_{c'}(x, y)} \right)$
 - Reminder: $H \in [0, \log_2 |C|]$
 3. Distribution consistency: $DC = 100 - \frac{1}{Z} \sum_{x, y} p(x, y) H(x, y)$
 - $Z = \sum_{x, y} p(x, y) \log_2 |C| / 100$



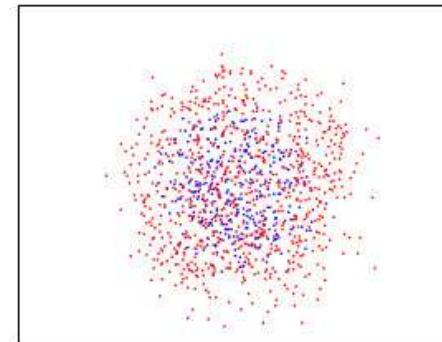
(a) 99



(b) 74



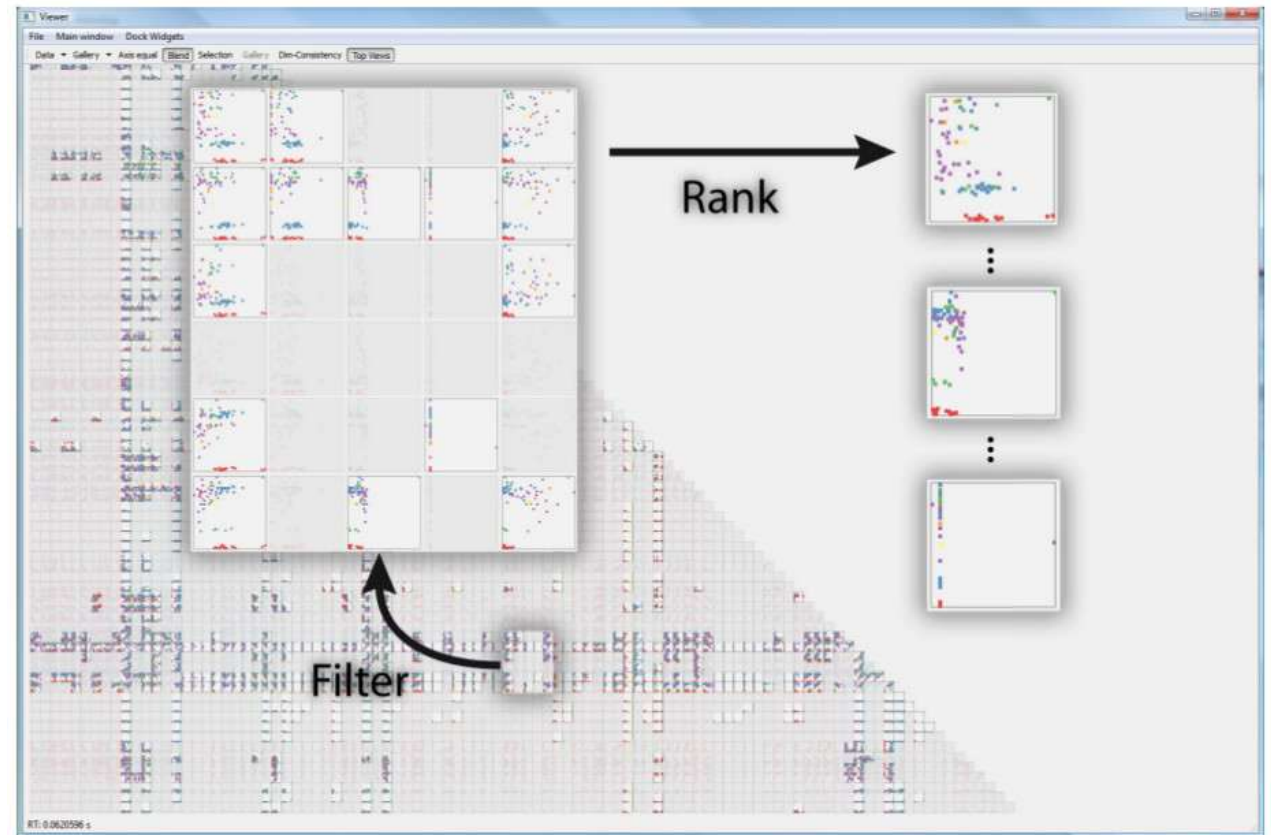
(c) 51



(d) 29

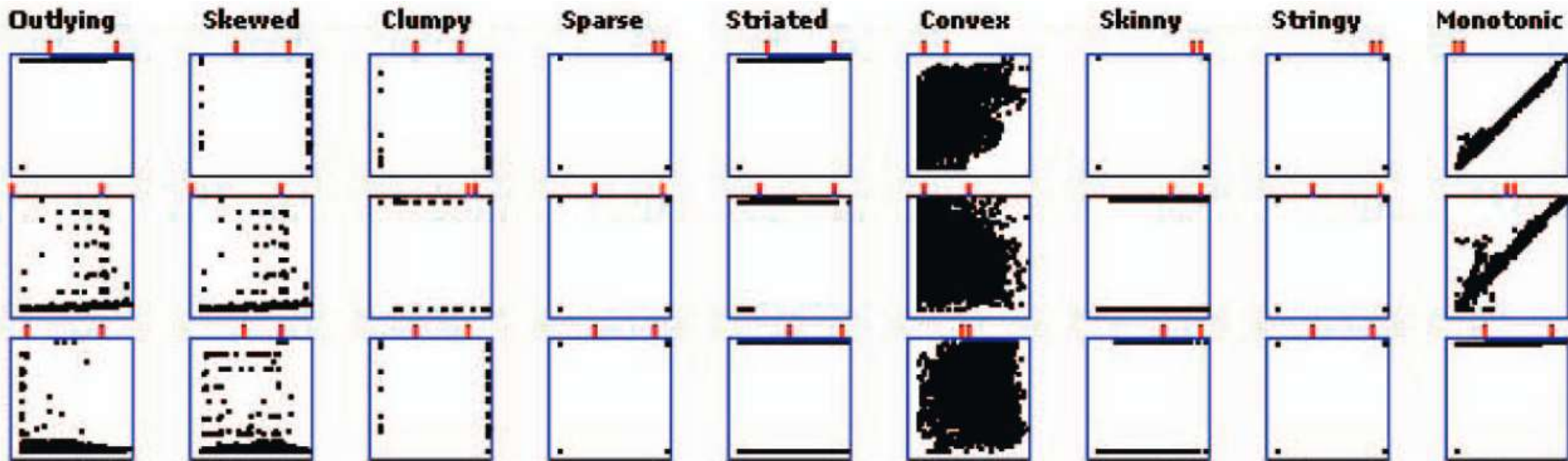
DC: WHO Example

- **WHO data**
 - 194 countries, 159 attributes, 6 HIV risk groups (>12,000 unique scatter plots)
 - Focusing on DC > 80 eliminates 97% of the plots
 - Highlighted rows: Single discriminative attributes
 - *Example:* Total health expenses



View Selection via Scagnostics

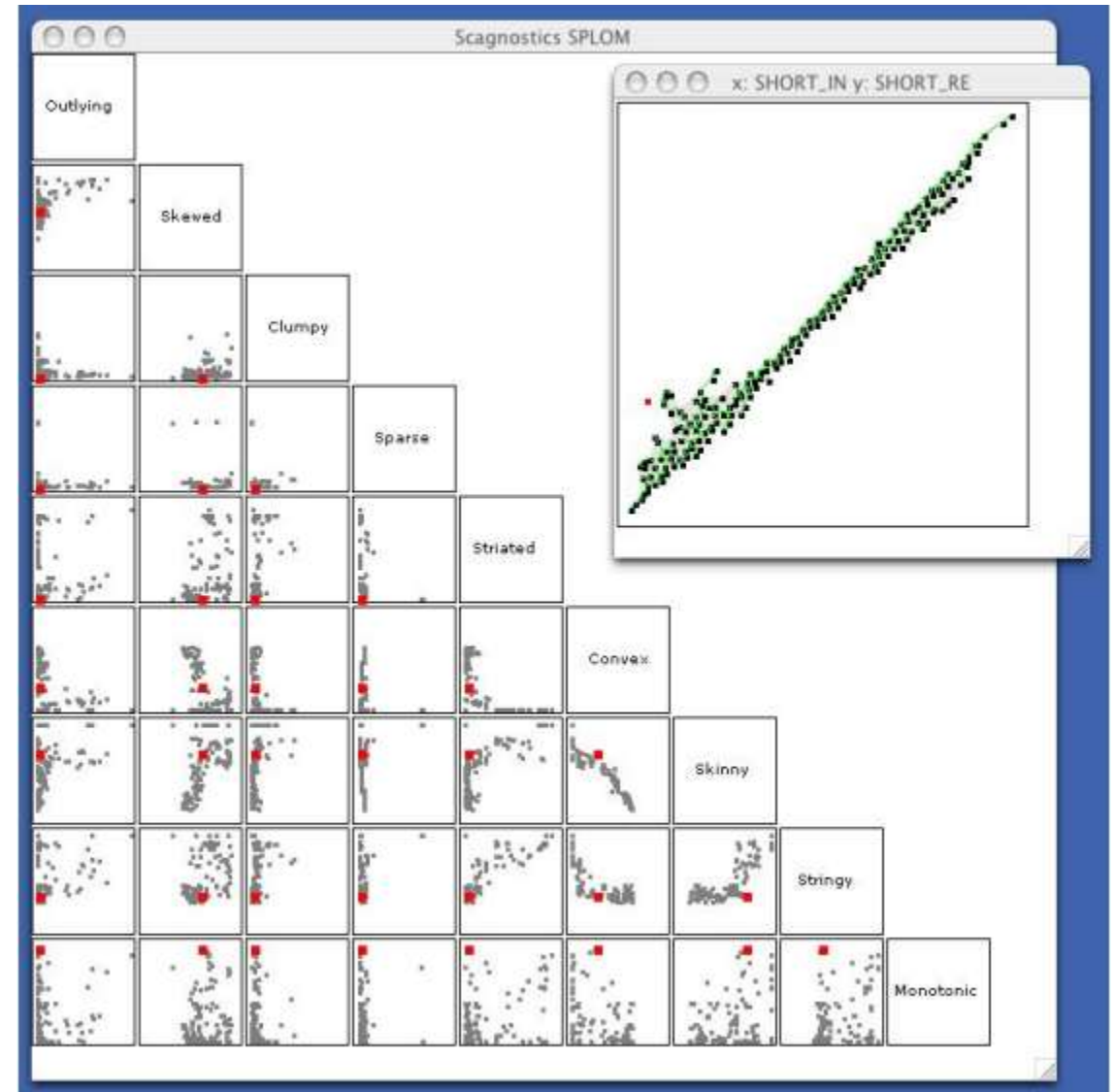
- *Idea:* Select scatterplots that exhibit specific patterns by characterizing them with **features** such as



- Term „Scagnostics“ („scatterplot diagnostics“) coined around 1985 by Tukey. A specific and computationally efficient definition of features was proposed by [Wilkinson et al. 2006]

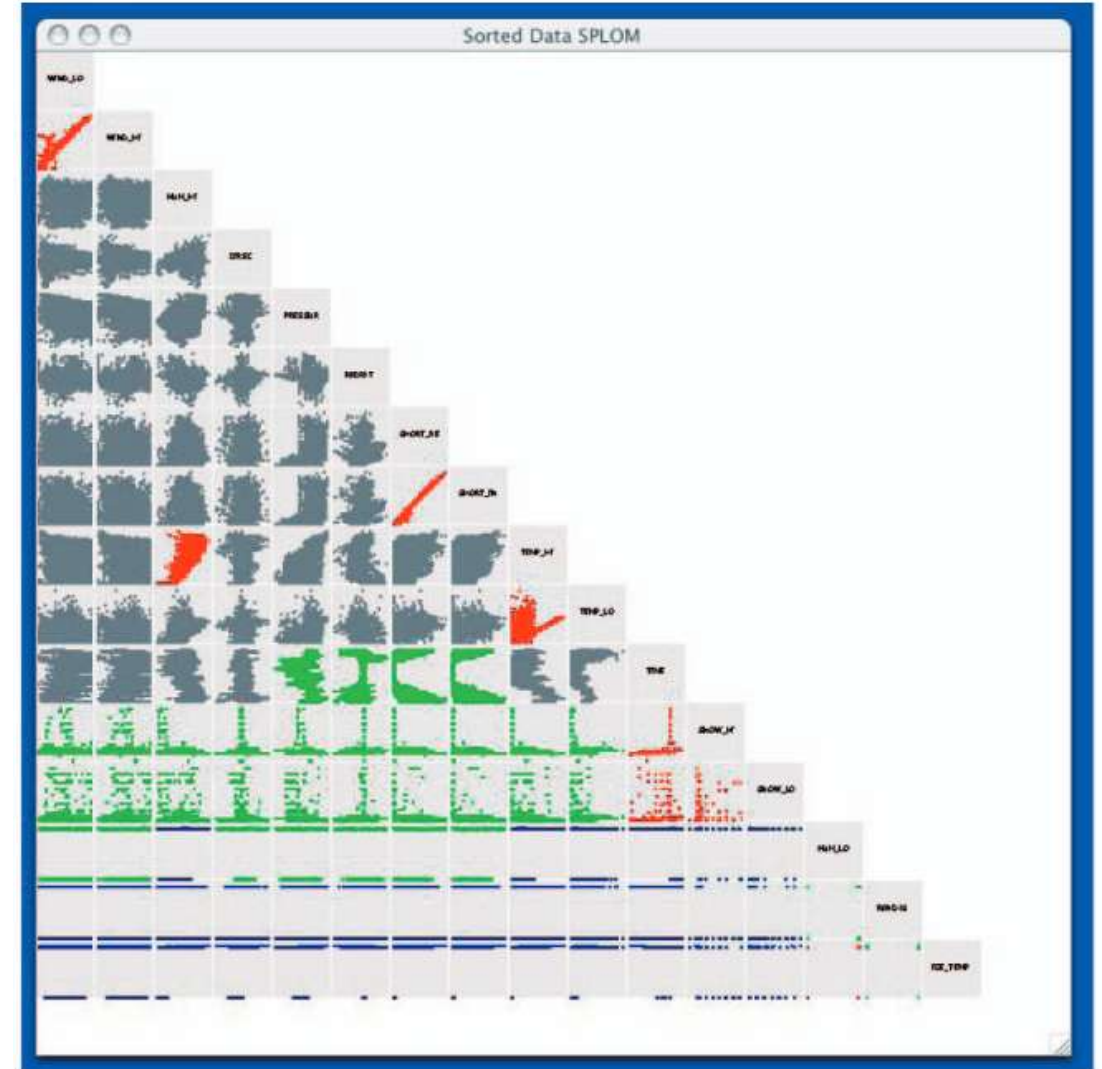
Features SPLOM

- Scagnostics **view selection** can be implemented with a second-level „feature“ SPLOM that links back to scatterplots of the original data



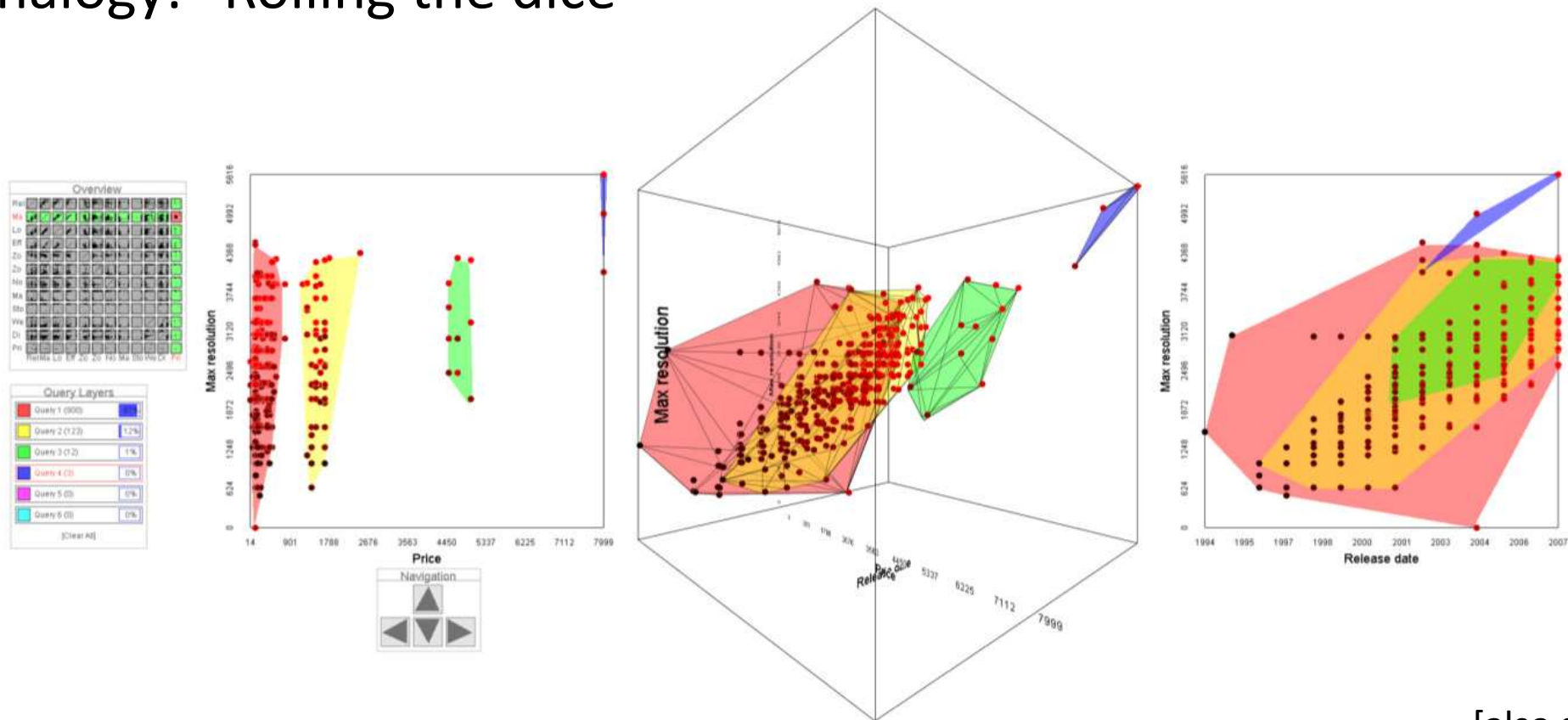
Sorting by Features

- Scagnostics features can also be used for **ordering SPLOMs** and for **clustering** plots in them
 - More comprehensive sorting criterion than Pearson correlation [Peng et al. 2004]
 - Sorting can be done by finding a direction in feature space along which the scatterplots differ most
 - One application of **Principal Component Analysis (PCA)**, which will be introduced in Chapter 3



SPLOM Navigation

- Animated 3D transitions between neighboring views
 - Analogy: “Rolling the dice”



Summary: Scatterplots

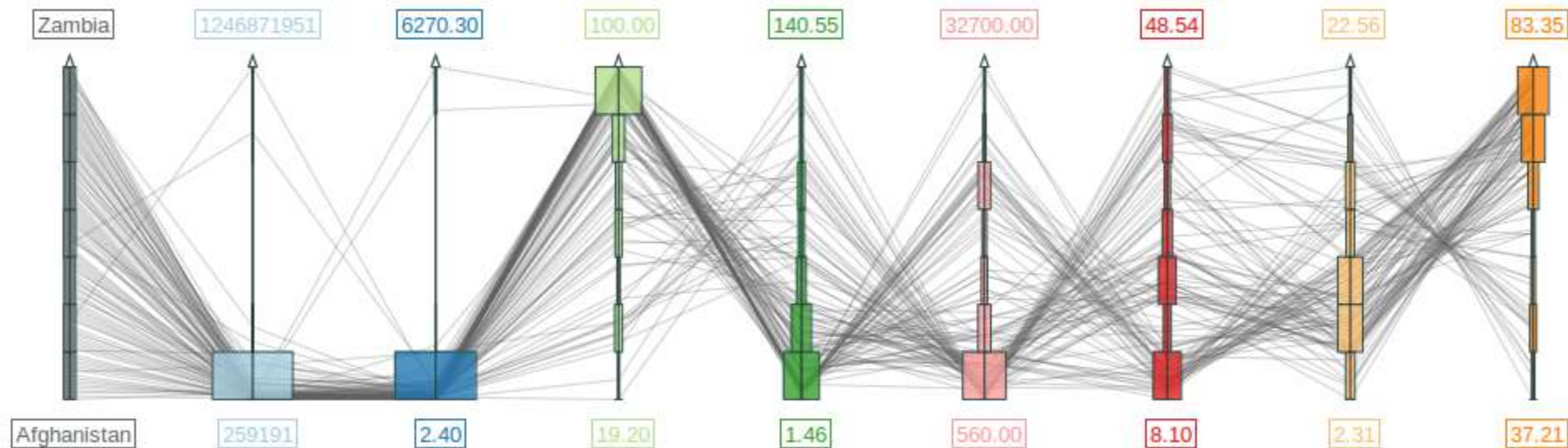
- **Scatterplots** are one of the most common techniques to visualize relationships between pairs of variables
 - Main limitation: Overdraw
 - Partial solution: Splatterplots
- **Scatterplot matrices (SPLOM)** generalize this technique to multi-dimensional data
 - Linking and animations convey correspondences between plots
 - Main limitation: Does not scale well to large number of attributes
 - Partial solutions: Sorting, view selection, scagnostics

Section 2.3: Parallel Coordinates

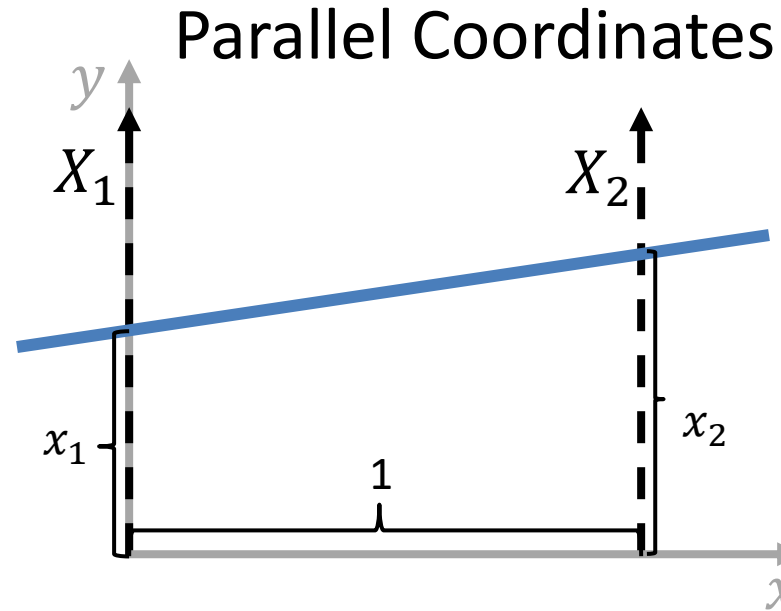
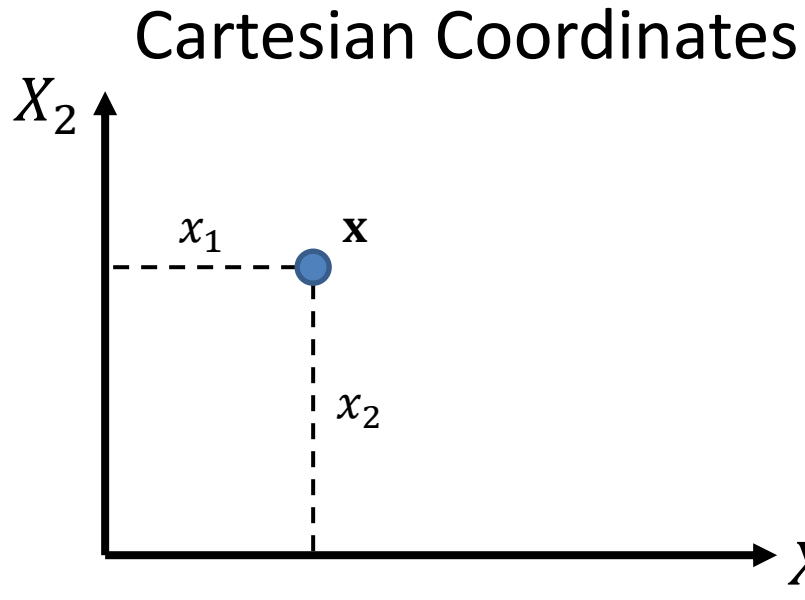
Parallel Coordinates

Parallel Coordinates can be used to visualize multi-dimensional data

- p equidistant axes which are parallel to one of the screen axes and correspond to the attributes
- The axes are scaled to the [min,max] - range of the corresponding attribute
- Every data item corresponds to a polygonal line which intersects each of the axes at the point which corresponds to the value for the attribute



Point-line Duality for Parallel Coordinates



- The point (x_1, x_2) in Cartesian coordinates is represented by the line $y = (x_2 - x_1)x + x_1$ in parallel coordinates
 - *Note:* We disregard the min-max axis scaling for this derivation
- It is easy to verify that all points on a line in Cartesian coordinates

$$x_2 = mx_1 + b$$

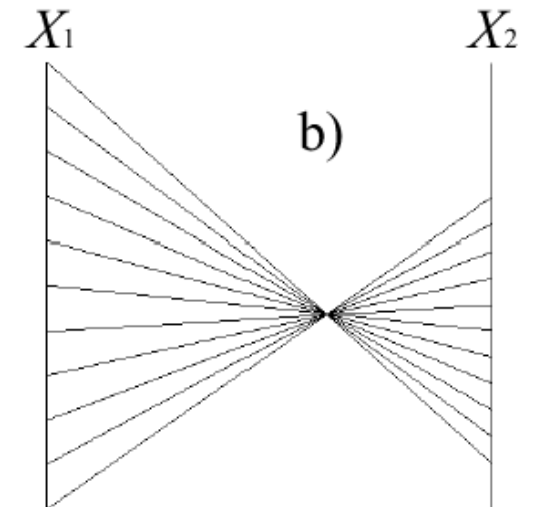
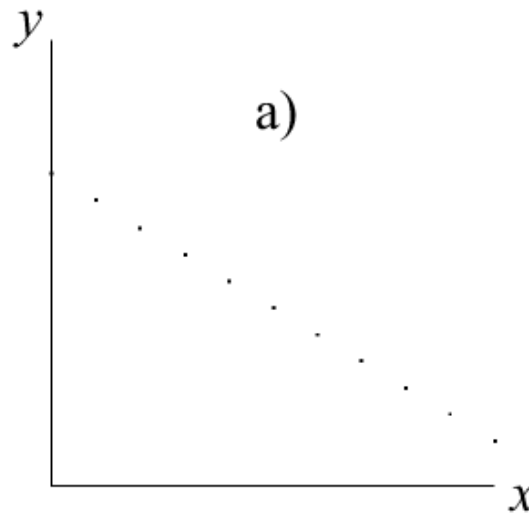
with $m \neq 1$ intersect in the point $\left(\frac{1}{1-m}, \frac{b}{1-m}\right)$ in parallel coordinates

Telling the Sign of the Correlation

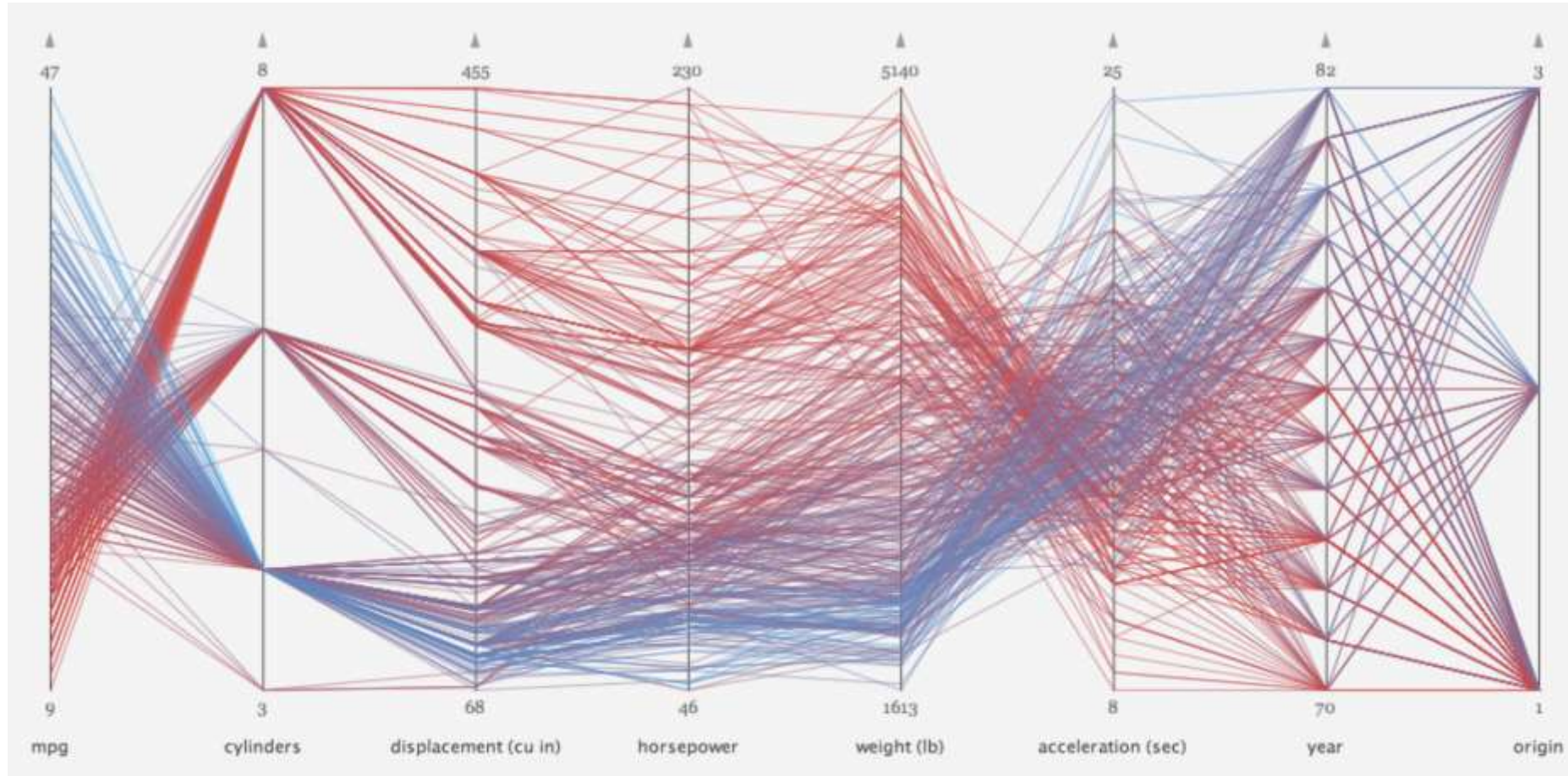
- *Reminder:* A straight line in Cartesian coordinates results in a single intersection point in Parallel Coordinates

$$\left(\frac{1}{1-m}, \frac{b}{1-m} \right)$$

- Between X_1 and X_2 for negative slope m
- Left of X_1 for slope $m > 1$
- Right of X_2 for $0 < m < 1$
- Parallel lines for $m = 1$



Example: Correlations in PCPs

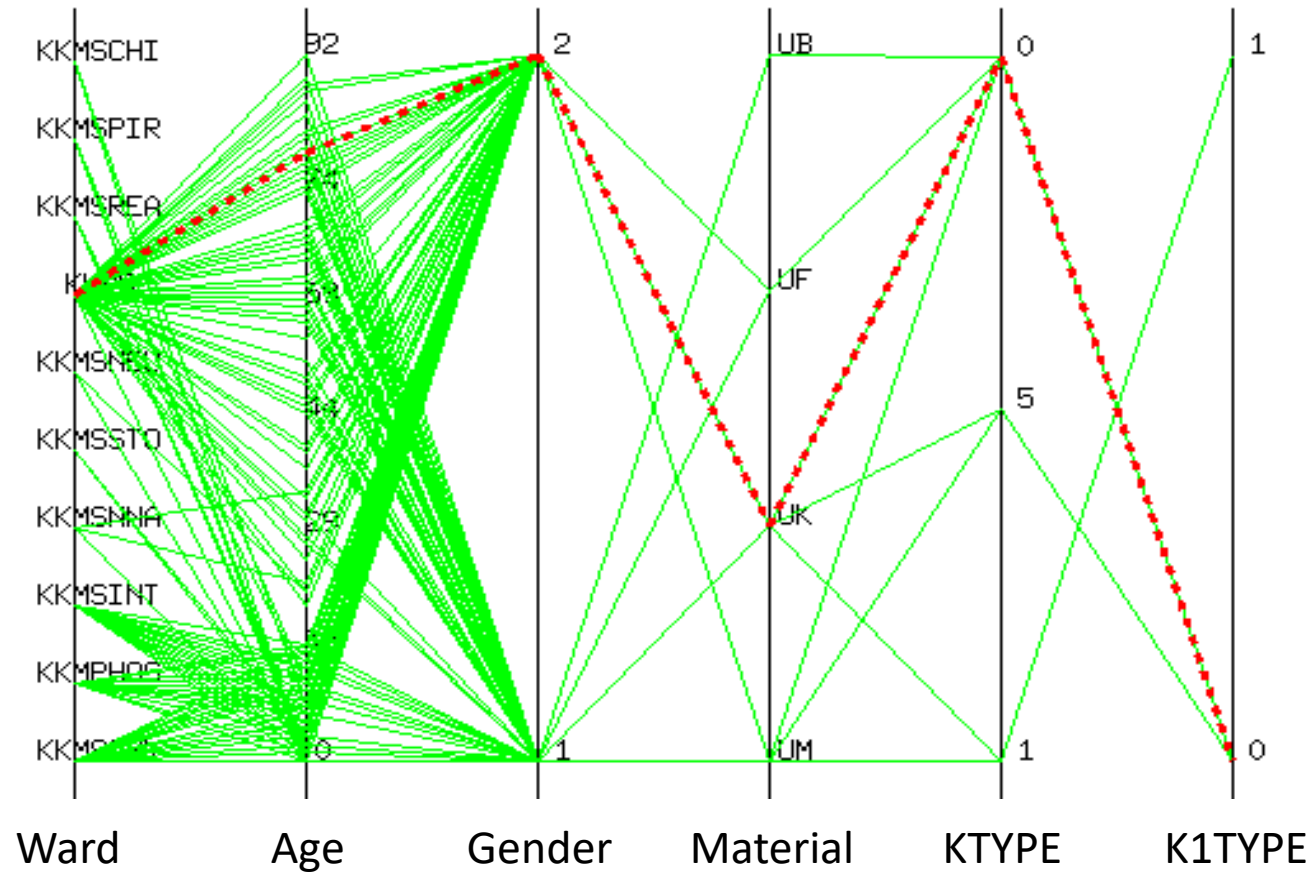


Quiz 1: Can you point out positive / negative correlations?

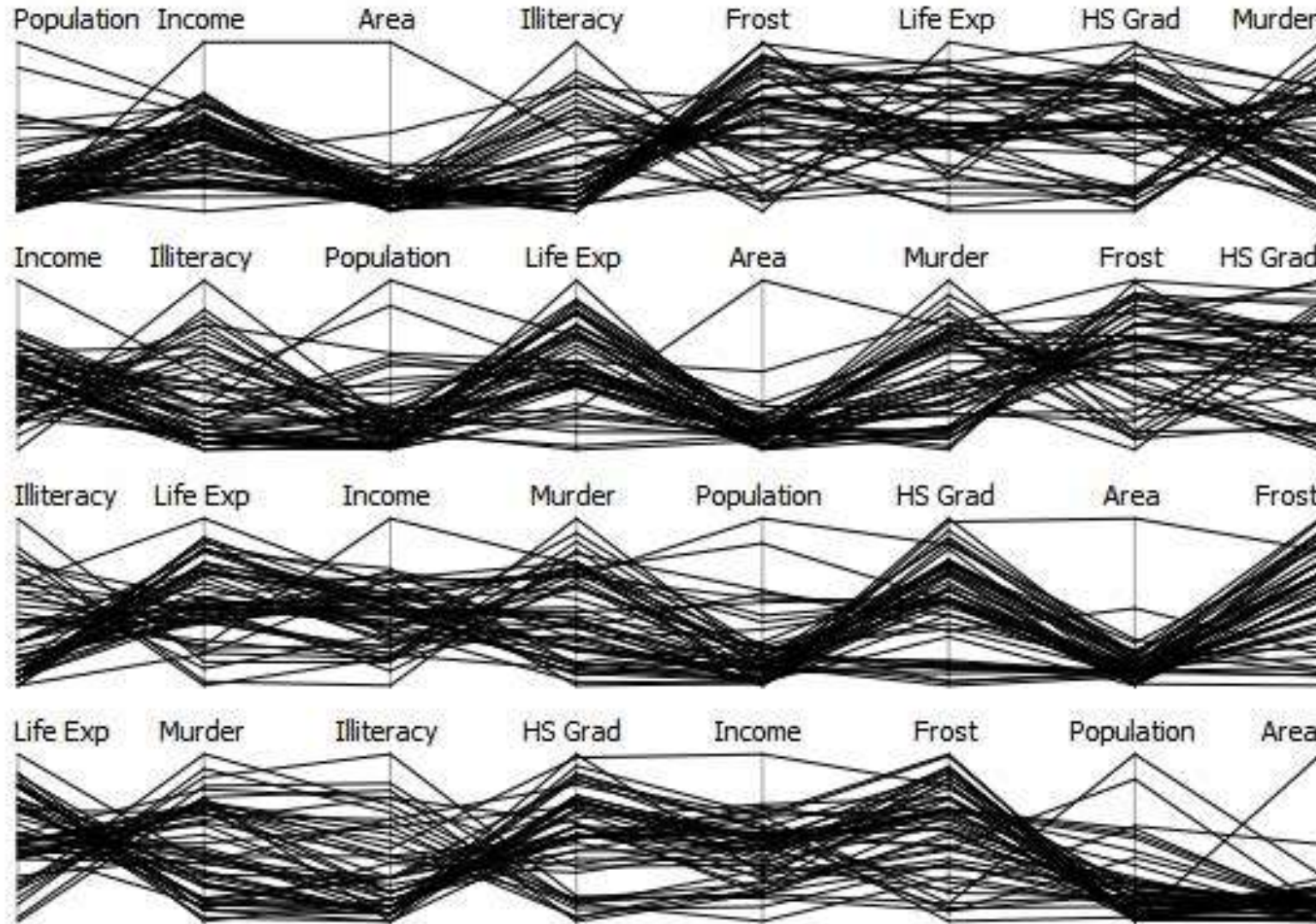
Quiz 2: Purpose of using color?

Example with Categorical Attributes

- Visualization of a hospital lab dataset:



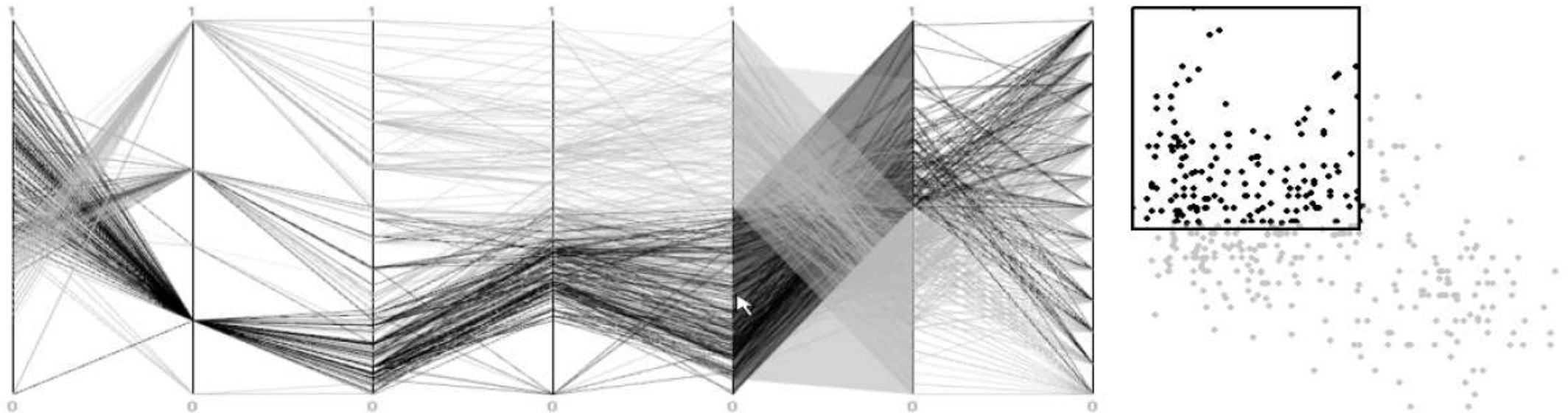
Axis Order Matters



Brushing and Linking

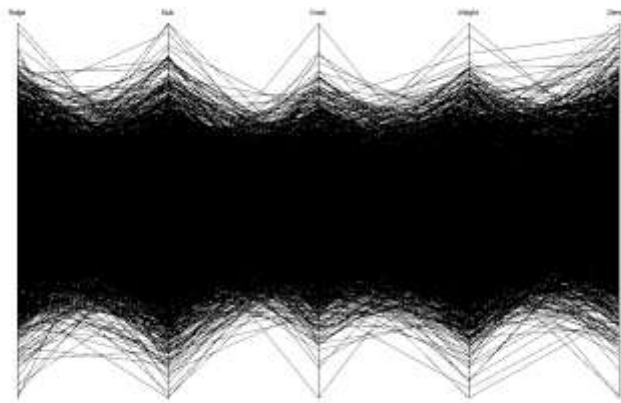
Interaction technique:

- “Brushing” = highlighting part of the data by selecting it with the mouse
- “Linking” = also highlighting the same data in another view

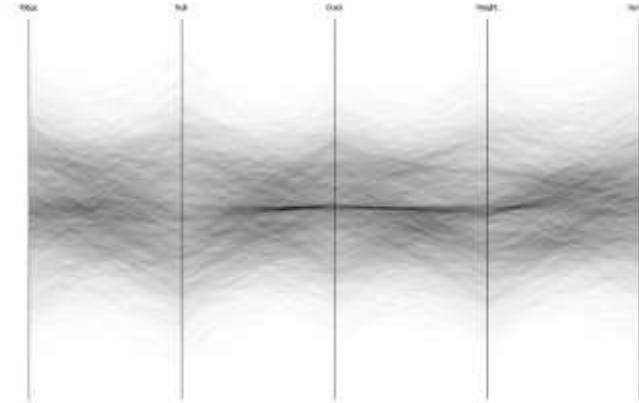


Overdraw in Parallel Coordinates

Parallel coordinates are even more susceptible to overdraw than scatterplots



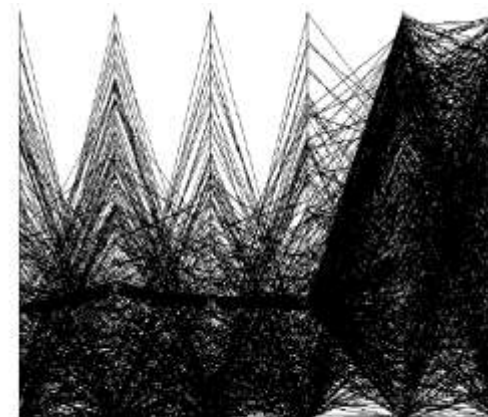
3848 data items (opaque)



Same data (semi-transparent)



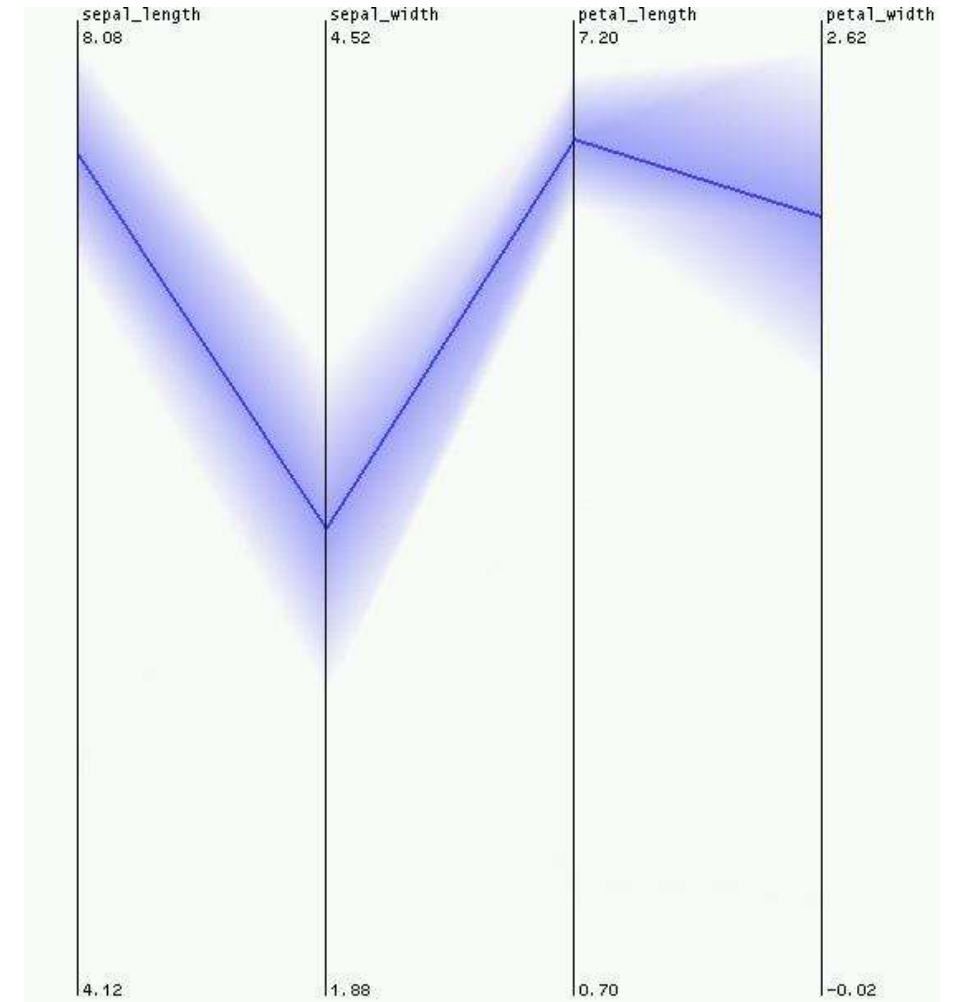
15,000 data items with noise



5% of the data (750 data items)

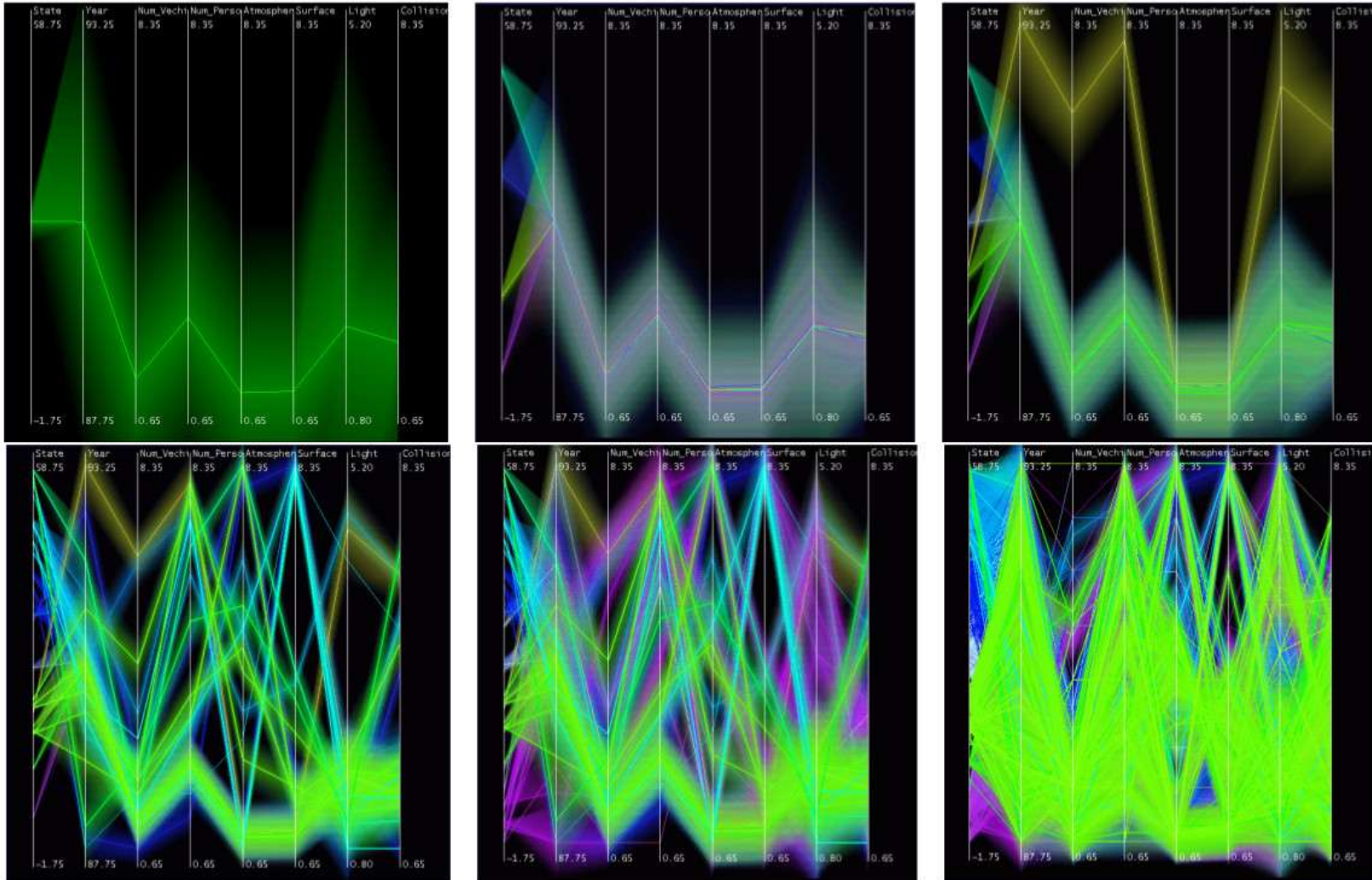
Hierarchical Parallel Coordinates

- Plotting **clusters** instead of individual data points is another way to reduce overdraw
- Strategy in [Fua et al. 1999]: Plot cluster as **graduated band**
 - Center = cluster mean
 - Width = cluster extent
 - Maximum opacity = cluster density



Example Result: Hierarchical PCP

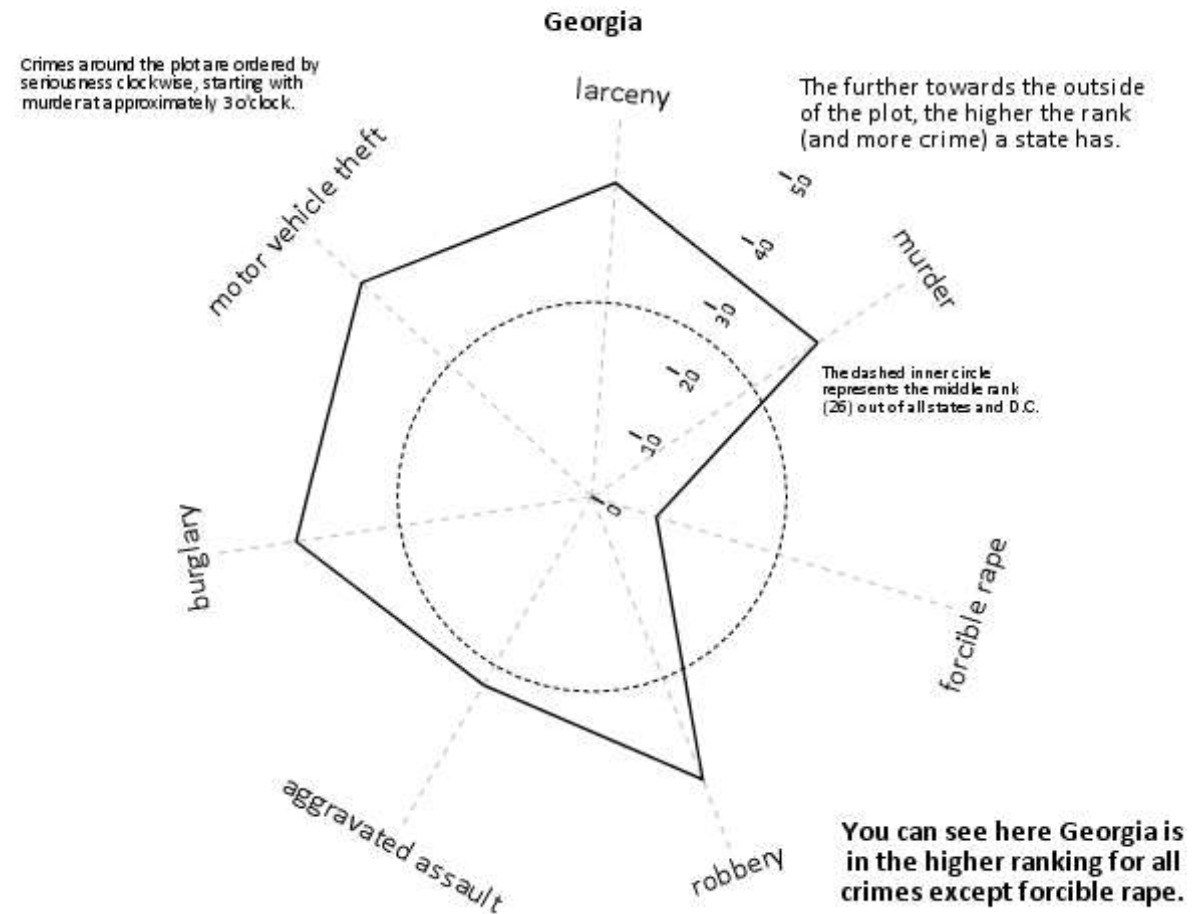
Exploration of clustering levels [Fua et al. 1999]



Radar Charts (aka Spider Charts, Star Charts)

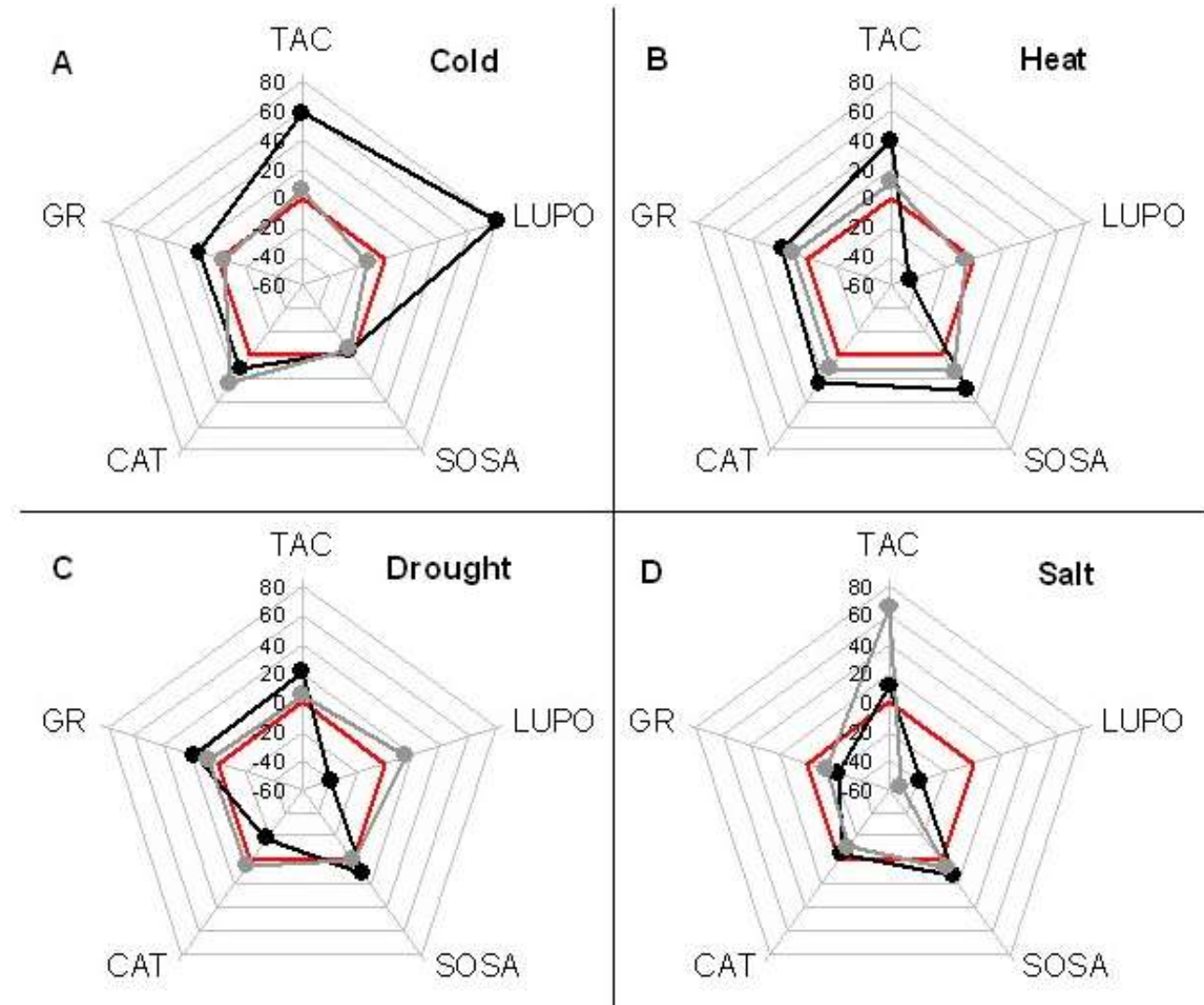
- **Radar charts** are similar to parallel coordinate plots, but with a radial axis layout
 - Axes are usually equally spaced with a common origin
 - Polygon intersecting each axis at the respective attribute value
- *Example:* Dotted circle shows median as a reference

Annotated Legend for Star Plots



Comparisons with Radar Charts (Star Charts)

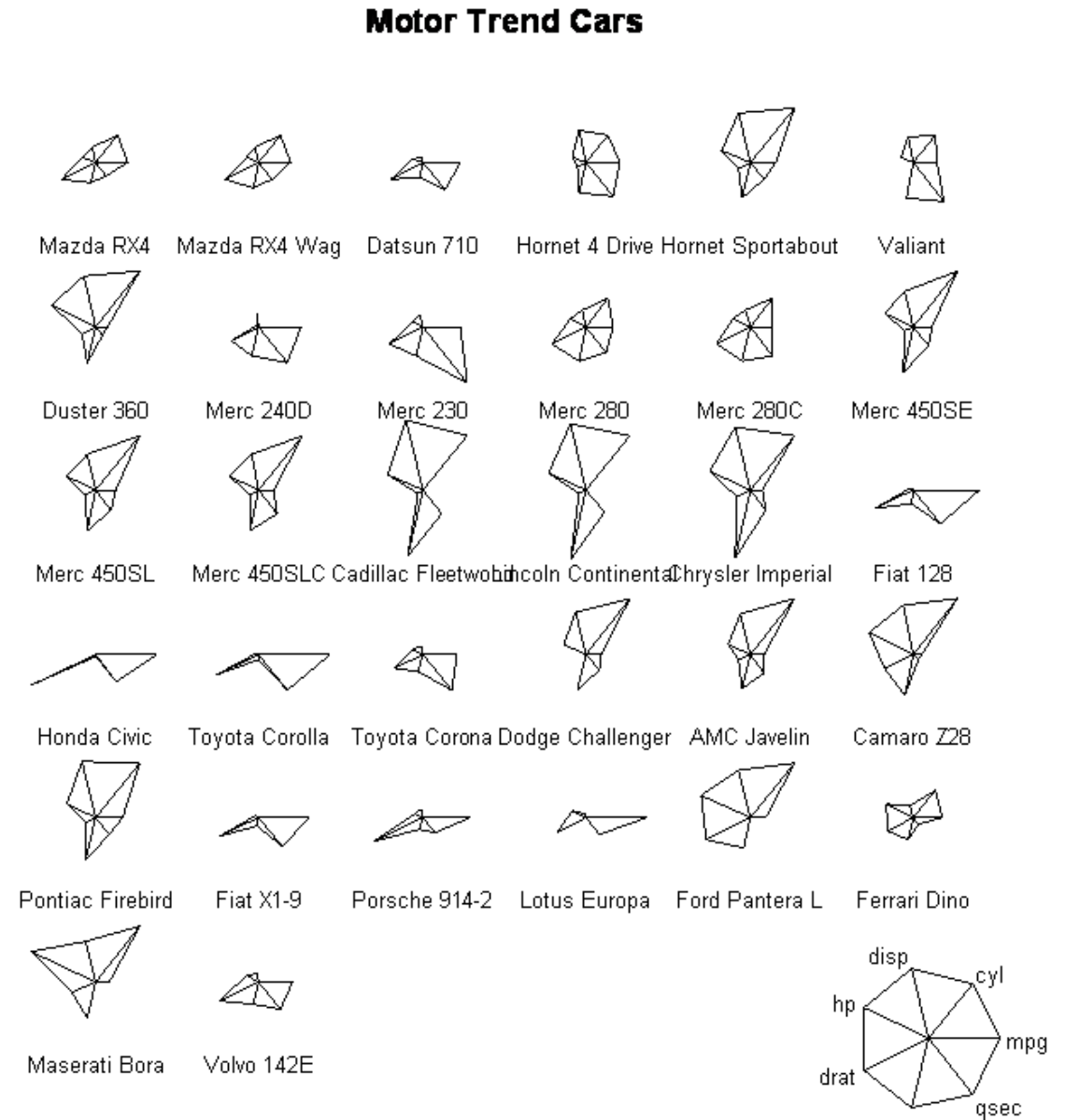
- Radar charts often overlay multiple data items in the same coordinate system for direct comparison
- *Example:* Effect of stress on the antioxidative system of plants
 - black: after stress
 - gray: after recovery
 - Different types of stress shown via faceting



[Saleh et al. 2009]

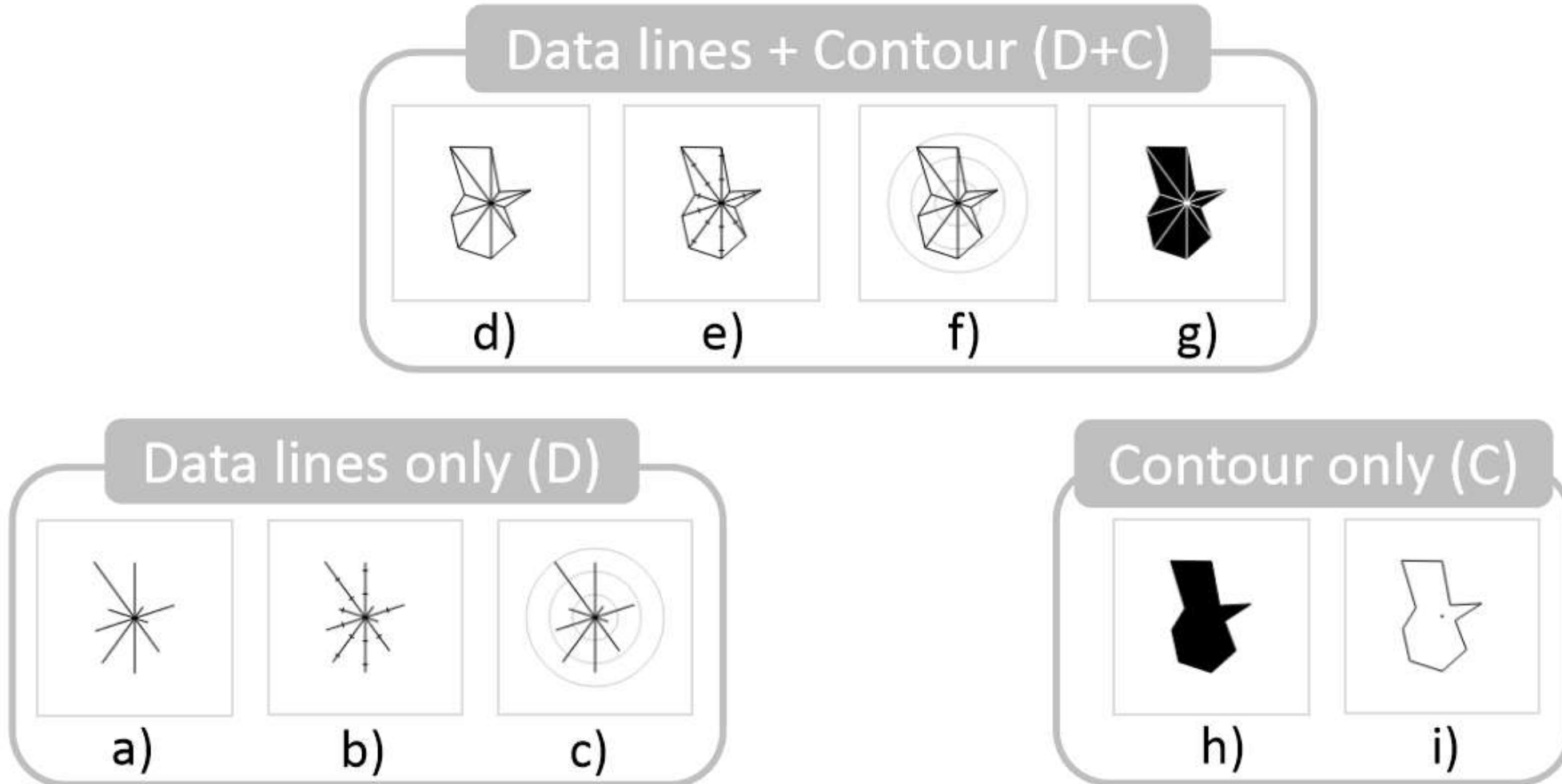
Star Glyphs

- Truncating the axes at the intersection points yields the **star glyph**
 - Emphasis is on the shape
- Star glyphs are often presented as **small multiples** to support finding data points with specific characteristics, similar data points, or outliers



Variations of the Star Glyph

- Many variations of the star glyph are in use [Fuchs et al. 2014]:



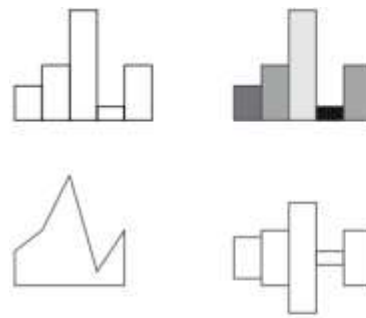
Summary: Parallel Coordinates

- **Parallel coordinates** are another standard method for multi-dimensional data
 - Unlike SPLOM, they visualize all dimensions in a single plot
 - Perfect correlations yield single intersection point
 - may not be in between the corresponding axes
 - may be “at infinity” (i.e., parallel lines)
 - Axis order and interaction (e.g., brushing / selection) are crucial!
 - Color can also help to reduce ambiguities and convey additional correlations
 - Even worse problems with overdraw compared to scatterplots
- *Variants:* Radar (spyder, star) charts, star glyph

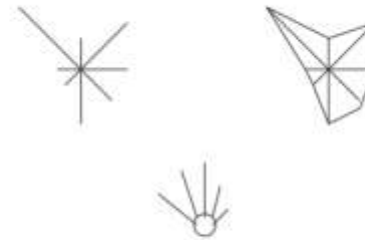
Section 2.4: Other Techniques

Glyphs and Icons

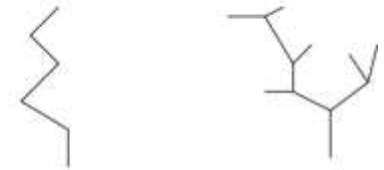
- **Glyphs** are small independent visual objects that are placed at discrete points in space and depict data attributes of an item
 - *Use visual channels such as:* Size, shape, style (e.g., dashed or dotted lines), orientation, color, texture, opacity, dynamics
- This definition is very broad:



Variations on Profile glyphs



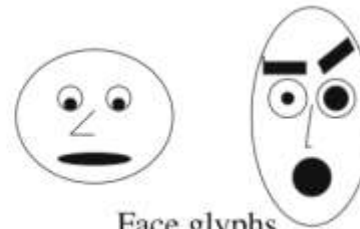
Stars and Anderson/metroglyphs



Sticks and Trees



Autoglyph and box glyph



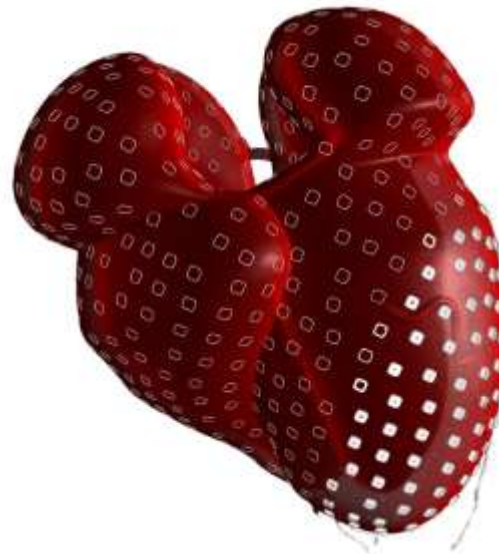
Face glyphs



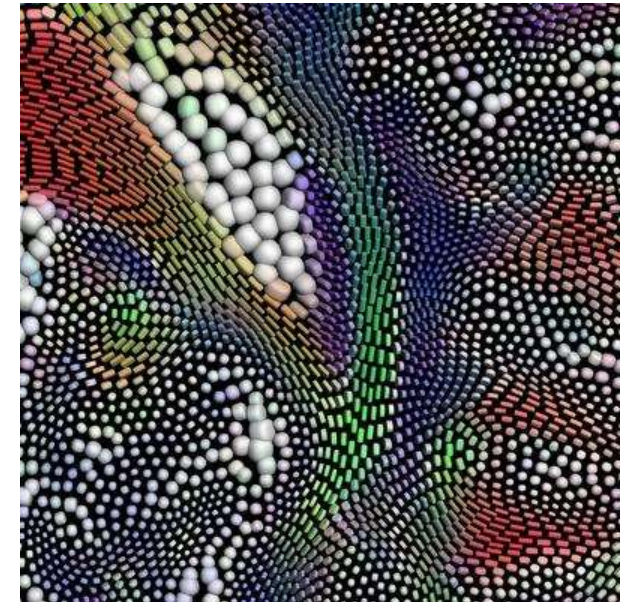
Arrows and Weathervanes

Glyphs: Design Considerations

- When **designing application-specific glyphs**,
 - Features should be easy to distinguish and combine
 - Icons with different values should be distinguishable
 - Placement needs to be carefully considered to avoid overlaps and visual clutter (esp. in 3D)
 - Feasible level of complexity will depend on density of glyph placement



[Ropinski et al. 2007]

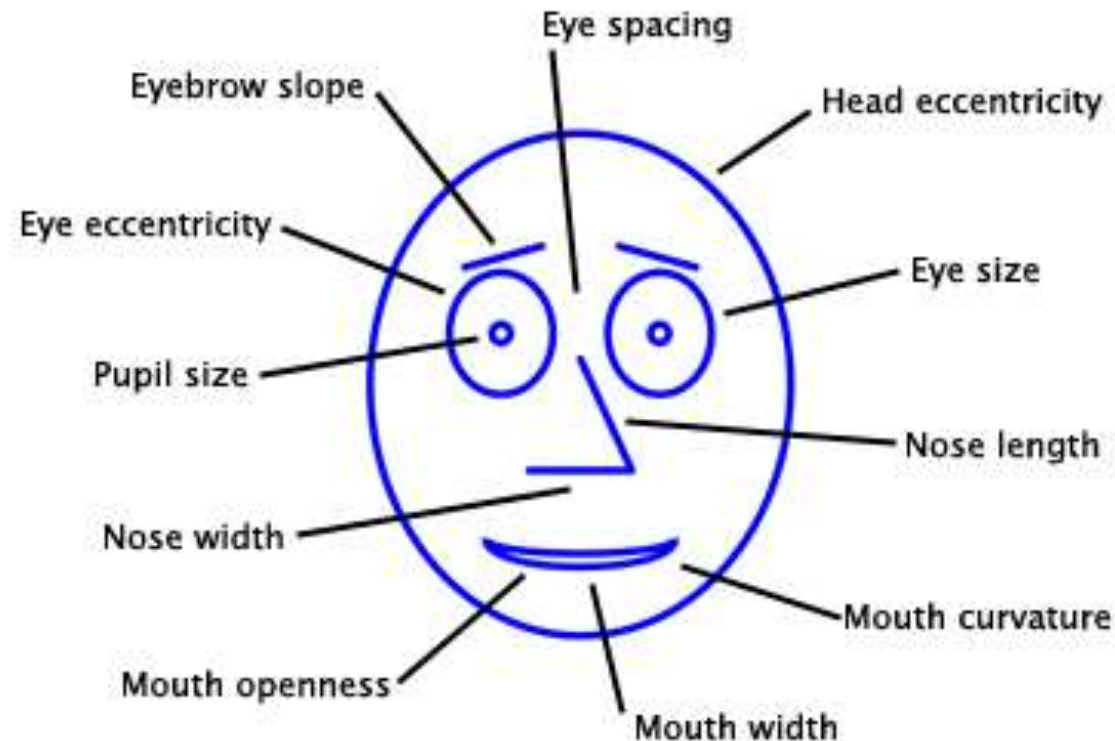


[Kindlmann et al. 2006]

Chernoff Faces

Proposed by [Chernoff 1973]

- Relies on human ability to distinguish small features in faces
- Each facial feature represents one variable



Example: Raw Fossil Data

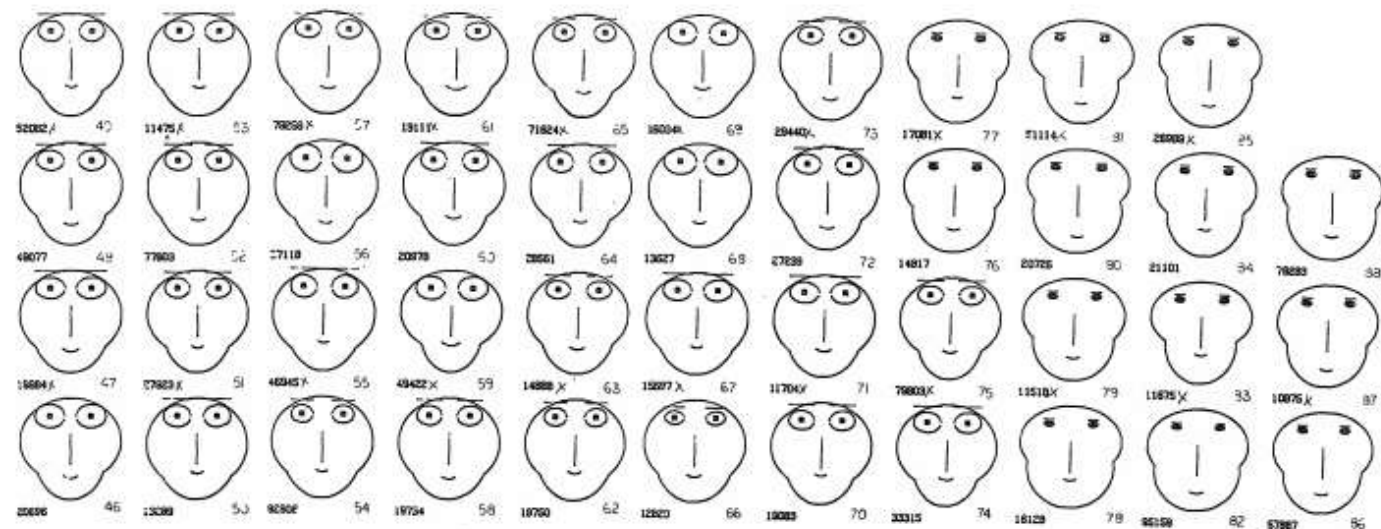
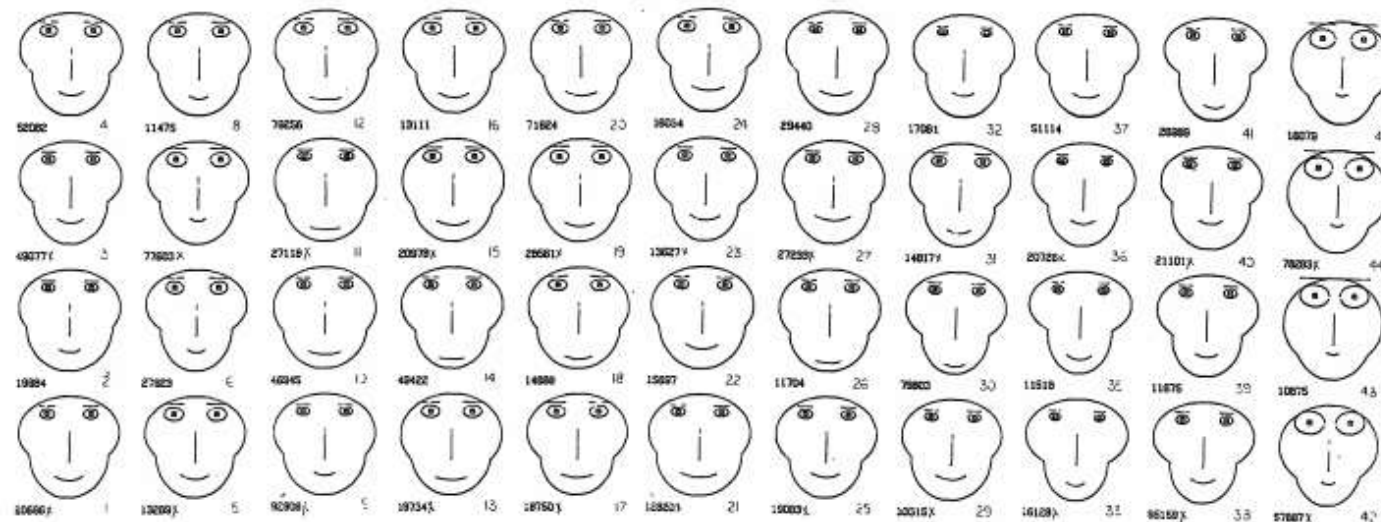
1'. 6 MEASUREMENTS ON 87 NUMMULITED SPECIMENS FROM THE EOCENE YELLOW LIMESTONE FORMATION, JAMAICA

ID	z ₁	z ₂	z ₃	z ₄	z ₅	z ₆	ID	z ₁	z ₂	z ₃	z ₄	z ₅	z ₆
1	160	51	10	28	70	450	48	190	34	9	26	96	1070
2	155	52	8	27	85	400	49	285	30	11	19	100	990
3	141	49	11	25	72	380	50	300	30	9	20	102	1120
4	130	50	10	26	75	560	51	225	30	10	22	105	985
5	161	50	10	27	70	665	52	260	34	8	22	97	1090
6	135	50	12	27	88	570	53	280	30	8	20	112	1200
7	165	50	11	23	95	675	54	300	34	10	20	108	835
8	150	50	9	29	90	580	55	310	30	11	19	106	1055
9	148	48	8	26	85	390	56	290	31	12	26	94	1240
10	150	45	7	31	60	435	57	260	30	8	22	98	1015
11	120	40	6	33	55	440	58	290	33	9	25	100	1010
12	120	51	8	32	56	650	59	160	31	11	20	79	1170
13	100	42	8	30	55	640	60	240	35	11	20	88	990
14	100	44	9	35	48	430	61	195	31	8	21	81	975
15	150	40	7	29	65	650	62	290	34	10	19	94	860
16	90	46	9	30	70	655	63	210	35	9	22	96	950
17	75	42	8	28	60	640	64	180	30	11	22	97	990
18	120	47	7	35	67	645	65	205	29	11	23	90	805
19	200	43	9	30	62	660	66	215	34	8	21	100	700
20	120	41	8	28	63	530	67	270	31	8	20	111	1170
21	105	50	7	27	64	435	68	290	30	9	23	102	1350
22	210	52	9	26	67	440	69	320	32	10	19	87	1160
23	90	40	10	25	68	430	70	210	30	9	18	112	1010
24	110	52	11	25	60	530	71	210	30	8	21	95	1190
25	100	43	9	25	70	440	72	185	34	9	25	96	1055
26	90	44	7	36	63	454	73	200	32	8	26	98	980
27	70	45	8	23	64	450	74	170	29	9	20	95	1095
28	100	48	9	27	65	355	75	140	30	9	20	98	990
29	130	52	9	25	70	380	76	90	52	8	24	120	210
30	90	45	11	37	74	350	77	110	49	9	22	130	220
31	80	46	10	32	78	450	78	100	56	8	19	128	216
32	95	49	10	25	82	260	79	95	49	8	24	124	218
33	70	44	12	30	85	262	80	65	62	9	30	134	200
35	95	51	15	31	70	270	81	55	50	10	27	128	205
36	100	46	11	24	76	270	82	70	53	7	28	118	204
37	95	48	10	27	74	355	83	85	49	11	19	117	206
38	85	47	12	25	73	360	84	115	50	10	21	122	198
39	70	48	11	26	78	365	85	110	57	9	26	125	230
40	80	54	10	21	80	370	86	95	48	8	27	114	228
41	85	55	13	33	81	355	87	95	49	8	29	118	240
42	200	34	10	24	98	1210	88	120	61	9	24	120	244
43	260	31	8	21	110	1220							
44	195	30	9	20	105	1130	min	70	29	6	18	48	198
45	195	32	9	19	110	1010	max	320	62	15	37	134	1350
46	220	33	10	24	95	1205	min ¹	70	40	6	21	48	260
47	220	30	8	25	90	1210	max ¹	210	55	15	37	95	675

¹ The second set of min and max refer to the ranges of the data over the first forty observations.

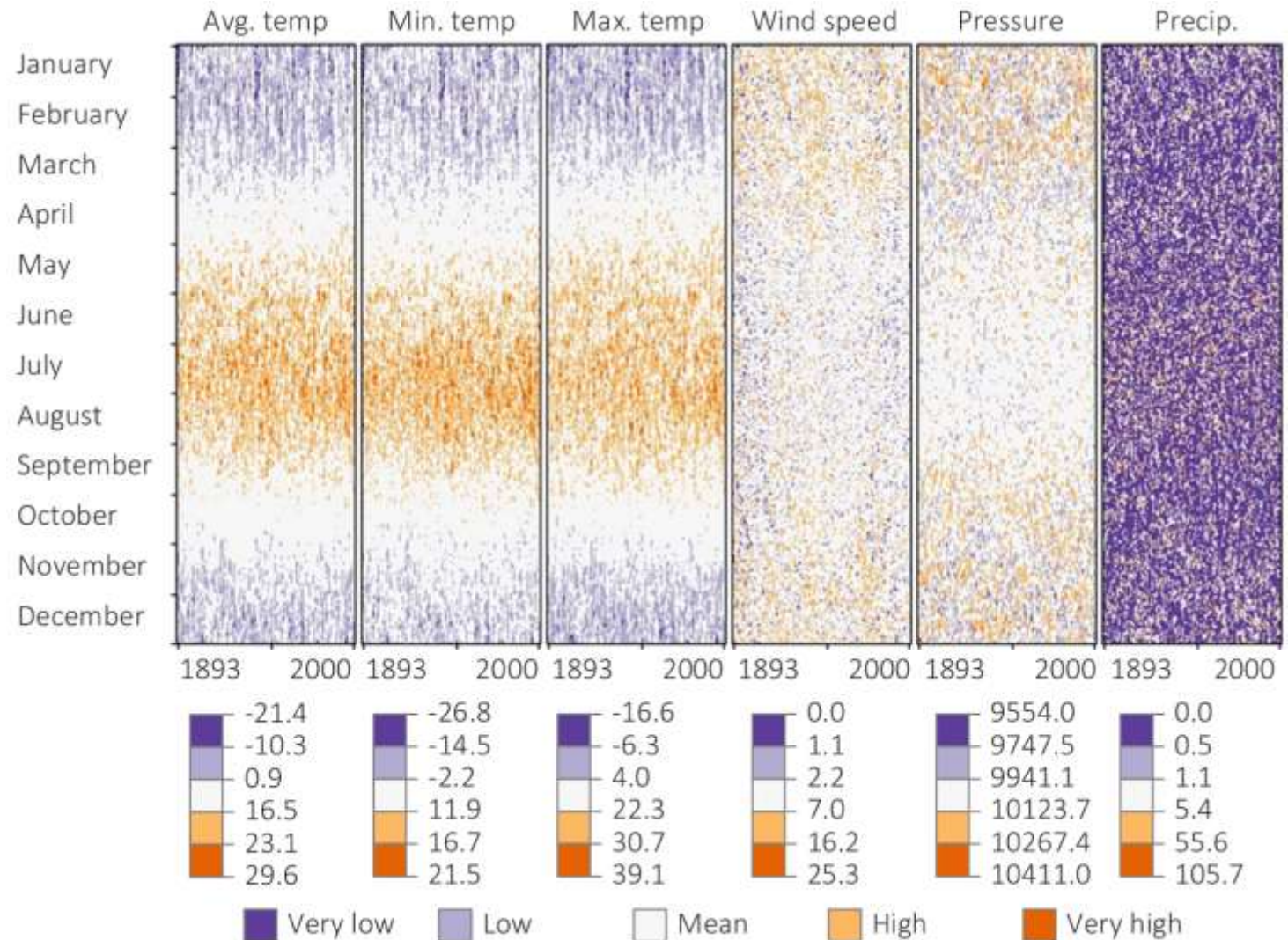
Example: Fossil Data as Faces

1A. FACES FOR 87 FOSSIL SPECIMENS OF EXAMPLE 1



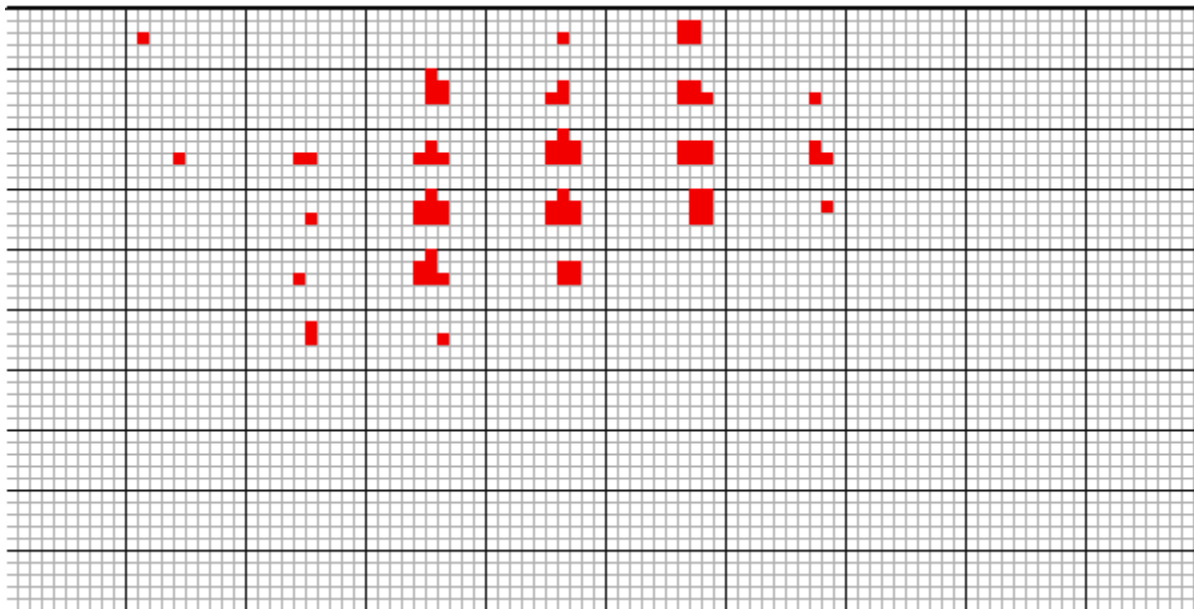
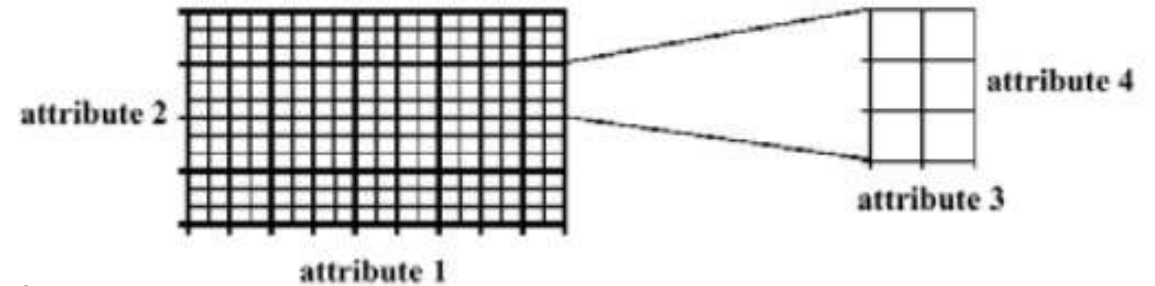
Pixel-Oriented Techniques

- *Idea:* Map each value to the color of a single pixel
 - Goal: Show as much detail as possible
 - Design: Decide on a suitable spatial layout and color code



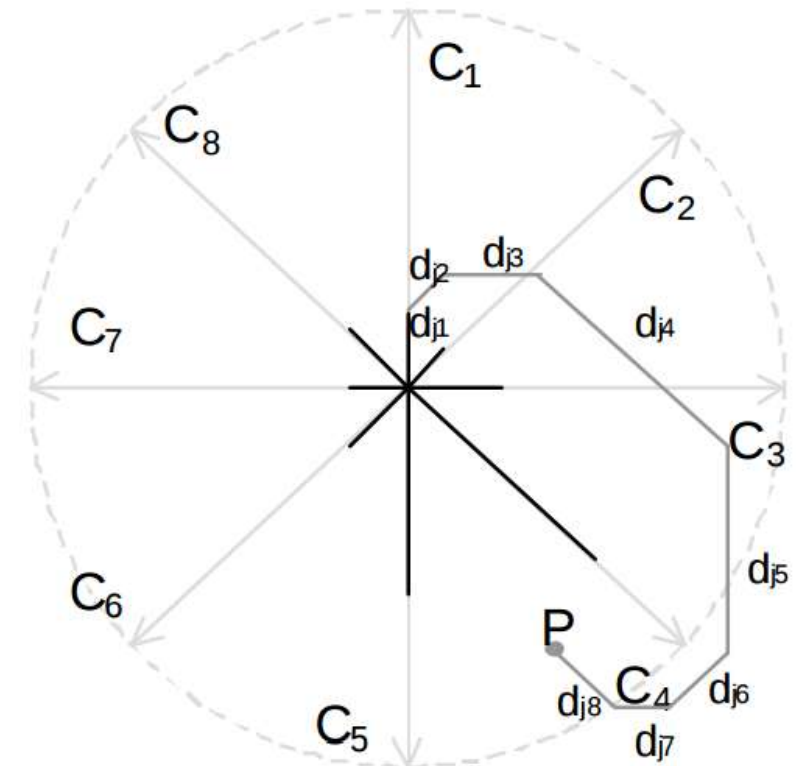
Dimensional Stacking

- Idea in [LeBlanc et al. 1990]: Recursively embed 2D histograms within each other
 - Bins of outer 2D histogram contain another 2D histogram, whose bins might contain yet another one...



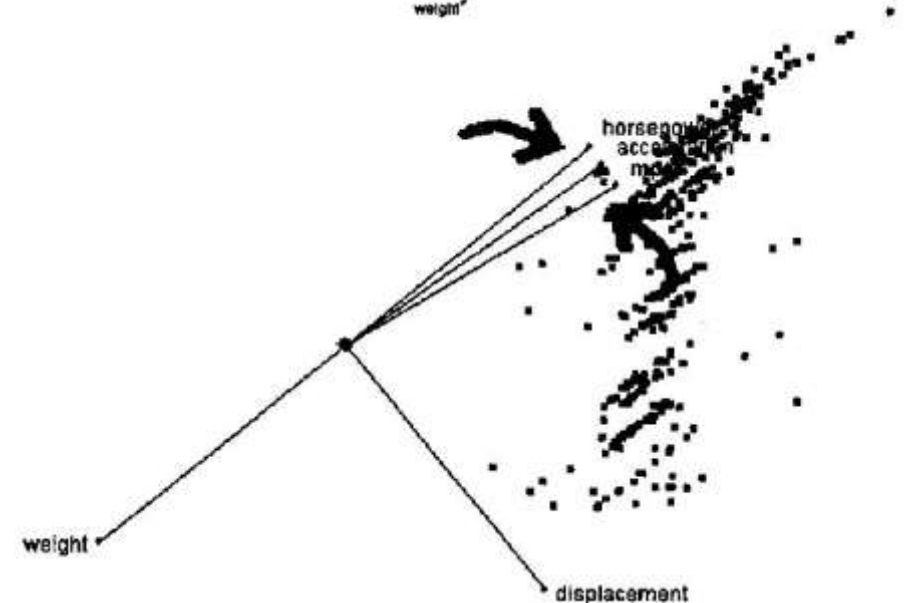
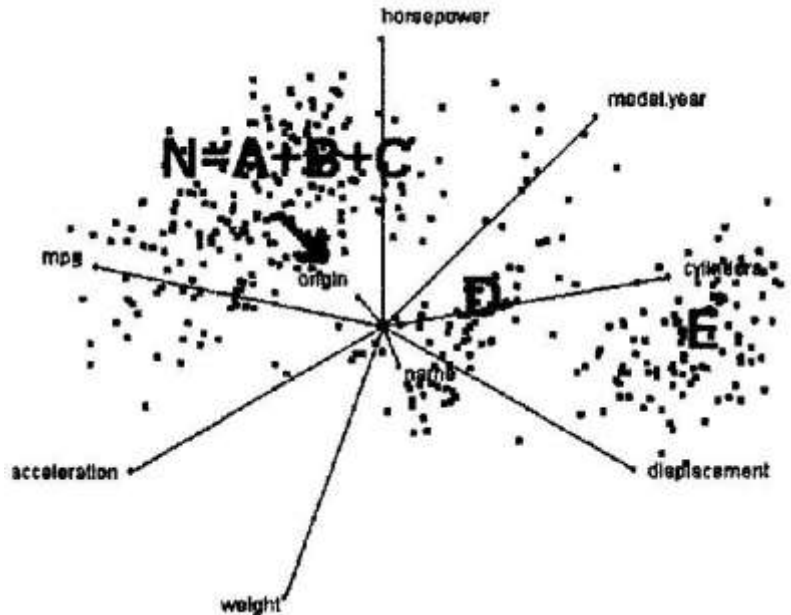
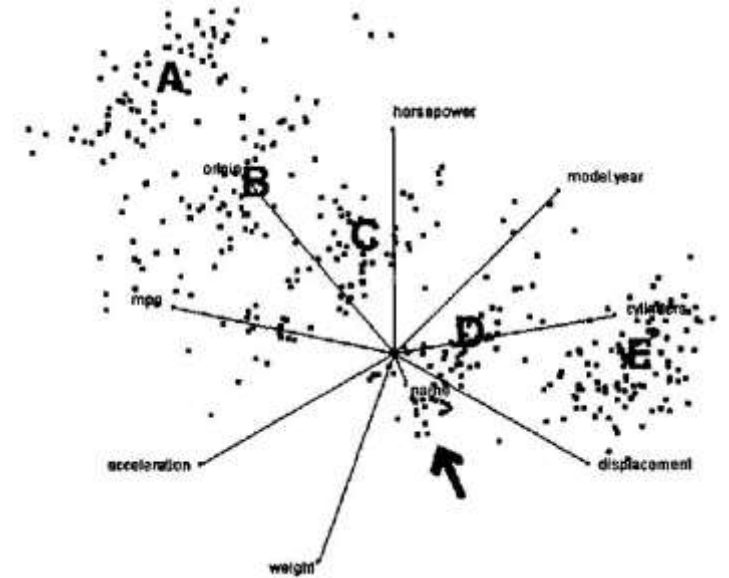
Star Coordinates

- Idea in [Kandogan 2001]: Arrange coordinate axes as a star in 2D
 - Unlike in star charts, each data item is represented as a point on the plane
 - Point is given by the vector sum of the scaled coordinate axes
 - Reveals cluster structure and allows for interactive exploration



Star Coordinates: Interaction

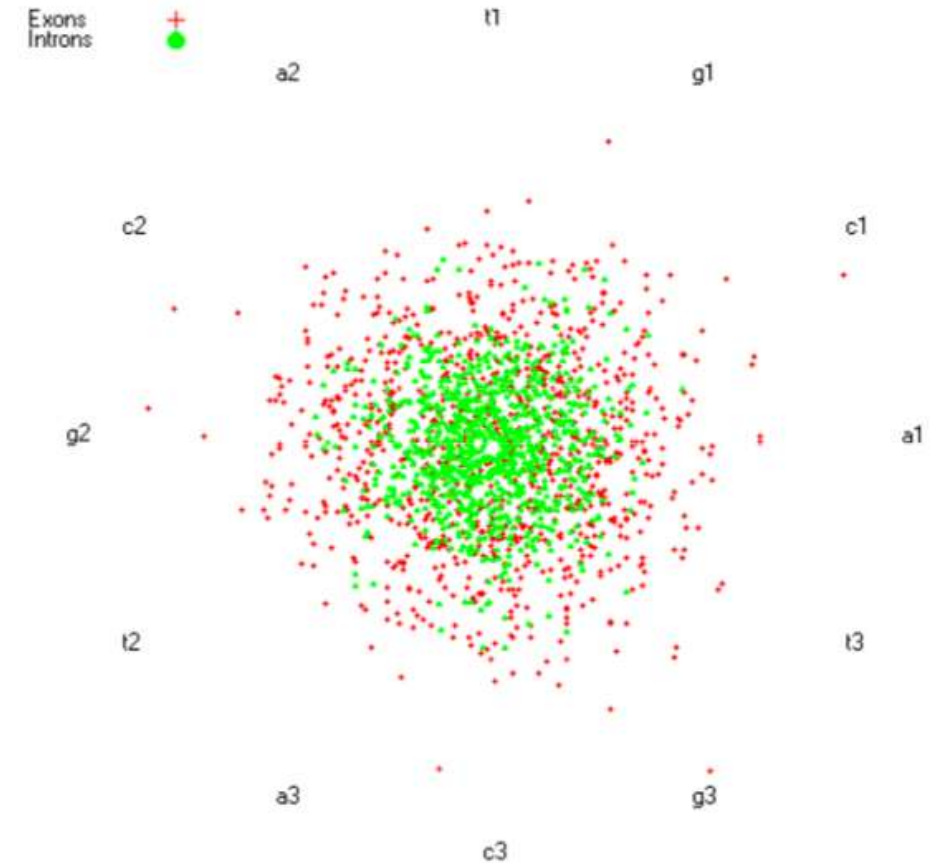
- Projection to 2D results in ambiguities that are resolved by interaction
 - Axis scaling, rotation
 - Brush points or axis ranges



RadViz

- **RadViz** is an alternative way to embed p -dimensional points $\mathbf{d}_i$ in two dimensions
 - Each dimension j represented by an **anchor point** $\mathbf{r}_j$, often uniformly distributed on circle
 - Attributes $d_{i,j}$ are min-max normalized and interpreted as **spring forces**
 - 2D embedding $\mathbf{x}_i$ given by equilibrium

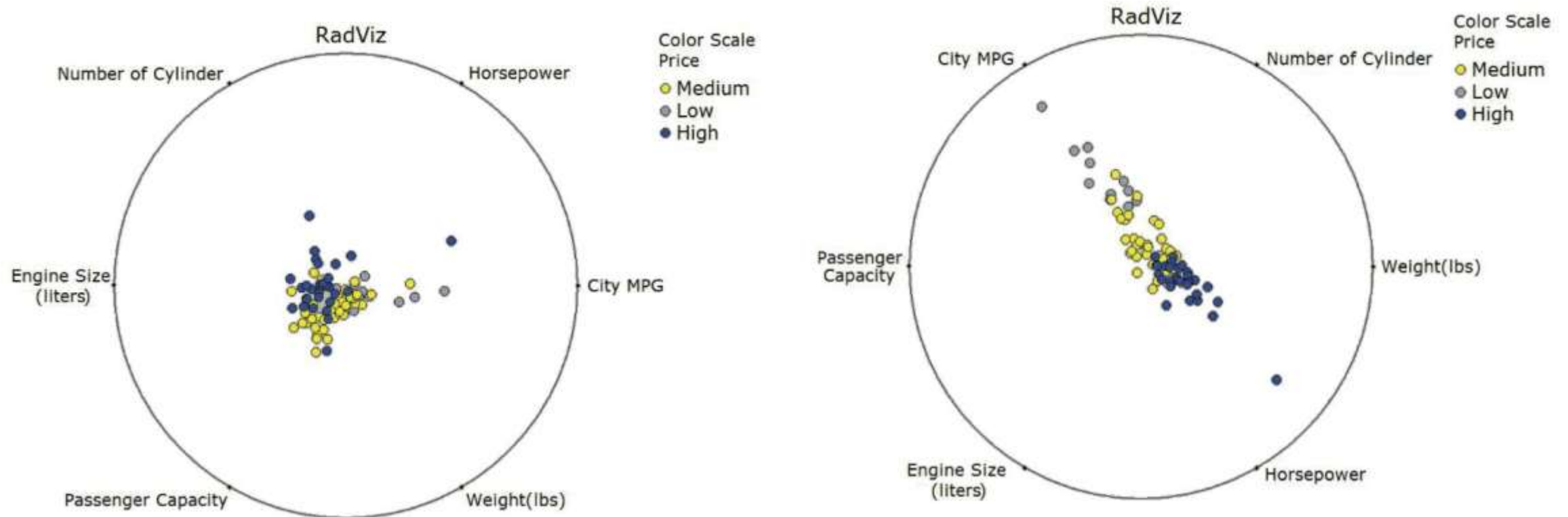
$$\sum_{j=1}^p (\mathbf{r}_j - \mathbf{x}_i) d_{i,j} = 0 \Rightarrow \mathbf{x}_i = \frac{\sum_{j=1}^p \mathbf{r}_j d_{i,j}}{\sum_{j=1}^p d_{i,j}}$$



[Hoffman et al. 1997]

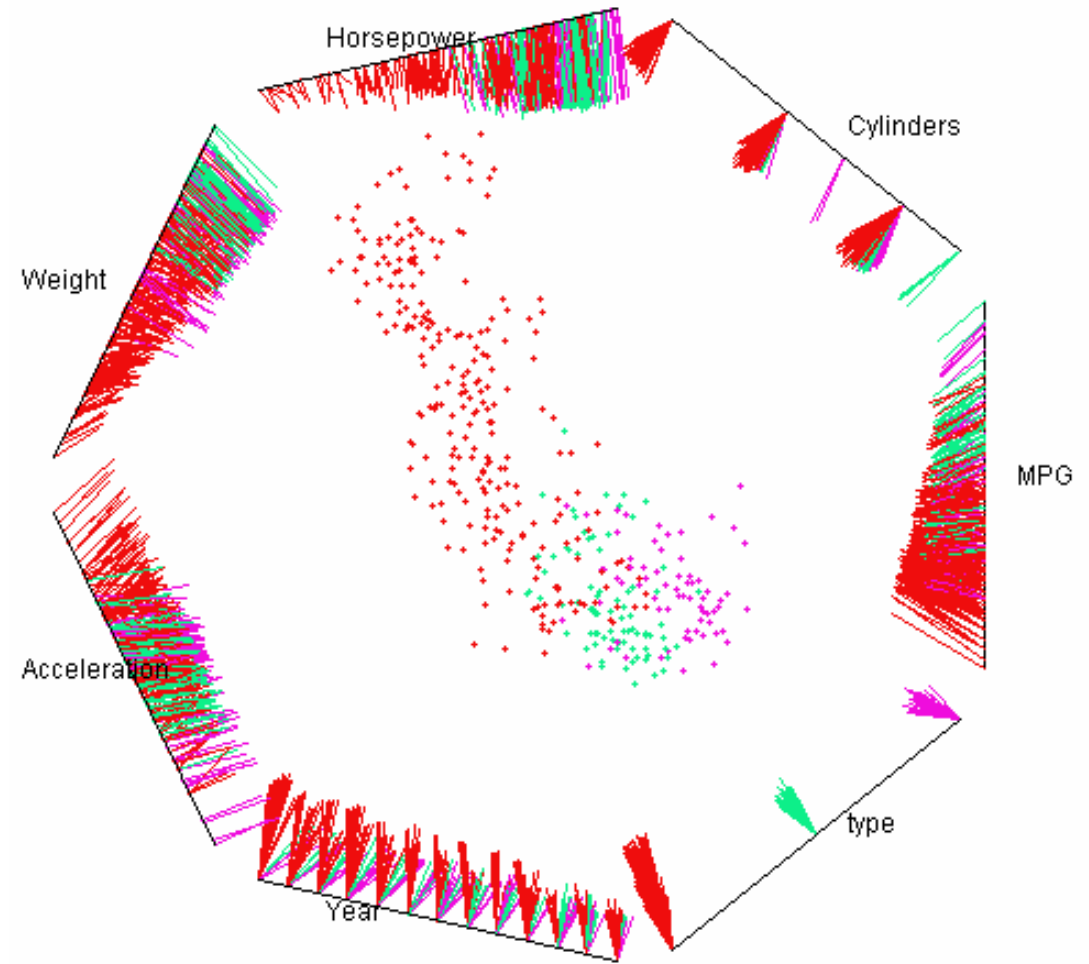
Interaction in RadViz

- Interactions such as **re-ordering anchor points** can help resolve ambiguities in RadViz



Extended RadViz

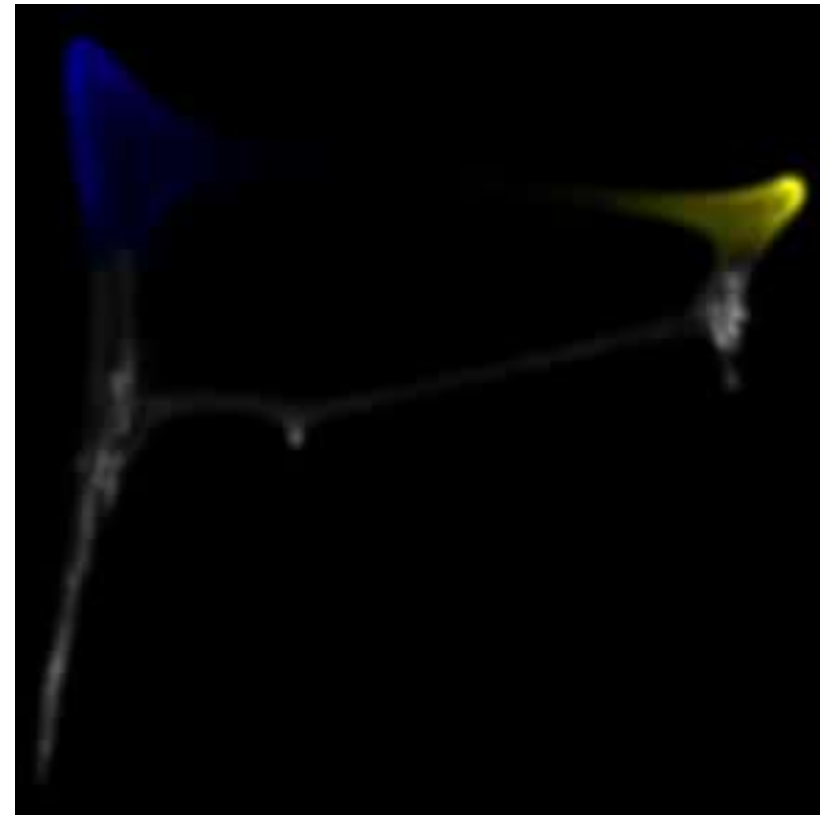
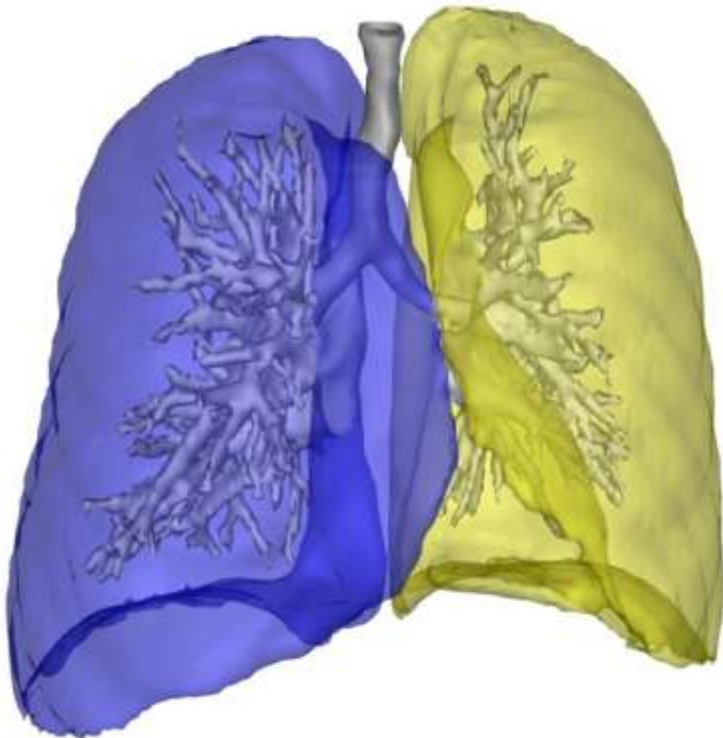
- **X-RadViz** aims to reduce overlap of points and increase interpretability by
 - spreading out anchor points along line segments
 - adding partial spring lines



[Hoffman et al. 1999]

Grand Tours

- *Idea* [Asimov 1985]: Once we have seen all possible 2D projections of a multi-dimensional space, we have seen the full space
- *Example*: „Grand tour“ of a 5-dimensional embedding that reflects distances on the surface (geodesic distances)

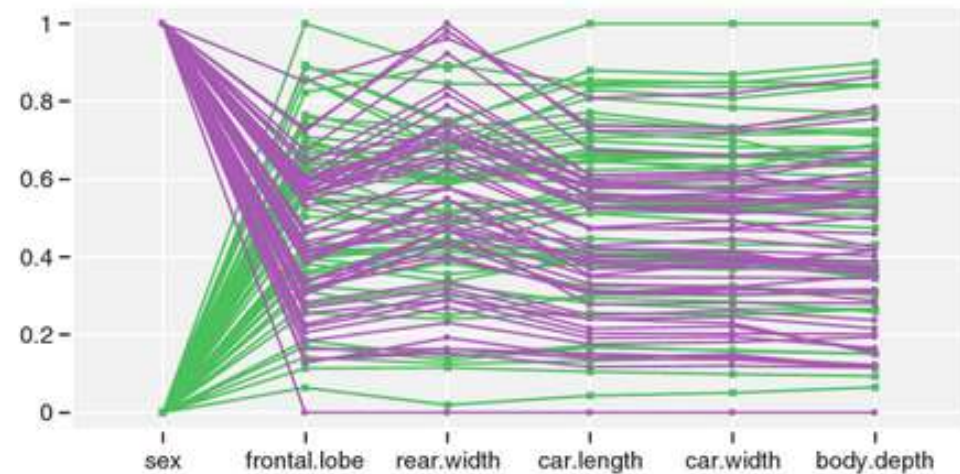
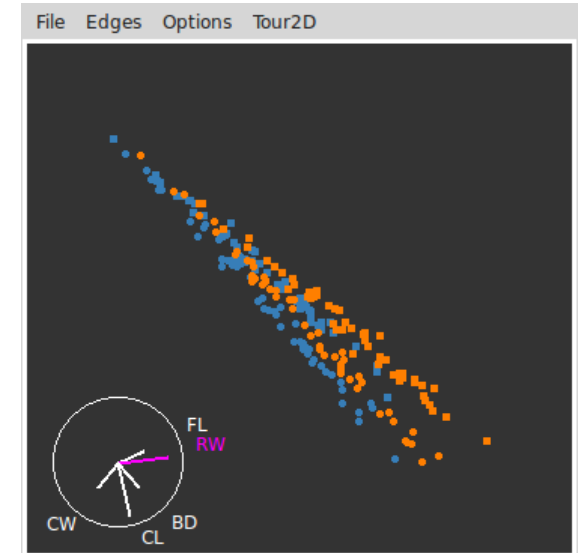


Grand Tours: Goals and Implementation

- A **grand tour** is a sequence of orthogonal projections (i.e., planes through the origin) that...
 - is **dense** in the space of all projections
 - For any possible projection, the tour contains a similar one
 - is **uniform** (free from bias, does not prefer specific views)
 - should move to new views **rapidly**...
 - ...while being **continuous** to allow for visual tracking
 - can be **reproduced** in later sessions
- *Possible implementation strategy:*
Interpolate between pseudo random projections
 - Store random seed to permit reproducibility

Grand Tour: Another Example

- From tutorial of open source software GGobi (ggobi.org)
- Dataset shows measurements of Australian crab, two species and two sexes each



Summary

- Other alternatives for encoding multi-dimensional data are
 - Glyphs
 - Pixel-oriented Techniques
 - Dimensional Stacking
 - Star coordinates (with interaction!)
 - (X-)RadViz
 - Grand Tours

Exam-like Questions

1. Suggest a way in which Kernel Density Estimation can be used for multidimensional data visualization.
2. Describe a strategy to support the navigation within Scatterplot Matrices (SPLOMs).
3. Name a limitation of Parallel Coordinate Plots and suggest a way to overcome it.
4. What is the difference between Star Glyphs and Star Coordinates?